



INDUSTRIAL AND SYSTEMS ENGINEERING PRINCIPLES & PRACTICE II

SEMESTER 2 AY2023/2024 (JANUARY 2024)

GROUP FINAL REPORT

From Traditional to Unmanned: Which Retail Checkout System is the Most Profitable?

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1. INTRODUCTION

1.1. Problem Description

In recent years, the retail landscape has witnessed a profound transformation through technological advancements. Among the notable innovations are novel check-out systems deployed in stores worldwide. Most commonly seen today are self-checkout kiosks, which still incorporate human assistance, albeit much less than a traditional checkout system. The rise of more avant-garde systems has become more apparent as well, pushing the boundaries of digitization and automation of checkout systems.

As brick-and-mortar stores continue to make up a majority of retail sales, retailers are seeking innovative ways to improve their customers' in-store shopping experience (MJV Team, 2024). Addressing this issue, technology has become the most favoured solution, with 47 per cent of retailers planning to implement a self-checkout system and 32 per cent of them planning to implement a checkout-free system in their stores (Zippin, 2023).

In light of the increasing interests of such systems, this project henceforth aims to conduct a comprehensive financial analysis to evaluate the viability and effectiveness of the different checkout systems based on their automation and affordability level.

1.2. Key Assumptions

1. Each alternative will be assessed by a **3-year study period**, corresponding to the duration of a typical commercial lease term in Singapore.
2. All alternatives follow the **co-termination assumption** where necessary.
3. All alternatives qualify for the **3-year Capital Allowance Scheme** under Singapore income tax law, and are subject to the **corporate tax rate of 17%**.
4. The MARR is **10%**, assuming the owner took a loan of 12% interest rate per annum to finance the shop's investment ($0.83 \times 12\% = 10\%$).
5. The **Market Value** of all alternatives will be **0** at the end of their service lives.
6. Customers' purchase from the shop will not be deterred by the type of checkout system.

7. As some costs are affected by the annual revenue, the following base, optimistic, and pessimistic annual values of retail store revenue will be used across all alternatives:

Optimistic	Base	Pessimistic
\$1,800,000.00	\$1,061,536.65	\$323,073.30

(Statista, 2018) (Baek, 2024)

2. DECISION ALTERNATIVES

In this project, three alternatives are considered - Conventional Checkout System, Four Self-checkout System, and Checkout-free System. In each alternative, a total of nine factors are included - Initial Investment, Initial One-off Grant Subsidy, Annual Sales Volume, Annual Benefit, Operations & Maintenance Cost (O&M) Cost, Manpower Cost, Shrinkage Cost, Market Value at EoY 3, and Useful Life.

For the breakdown of each value, please refer to the respective sheets in the MS Excel Workbook.

2.1. Alternative 1 – Conventional Checkout System

In this alternative, the scenario where the checkout process of a retail outlet is managed by two cashiers with two traditional checkout systems is evaluated.

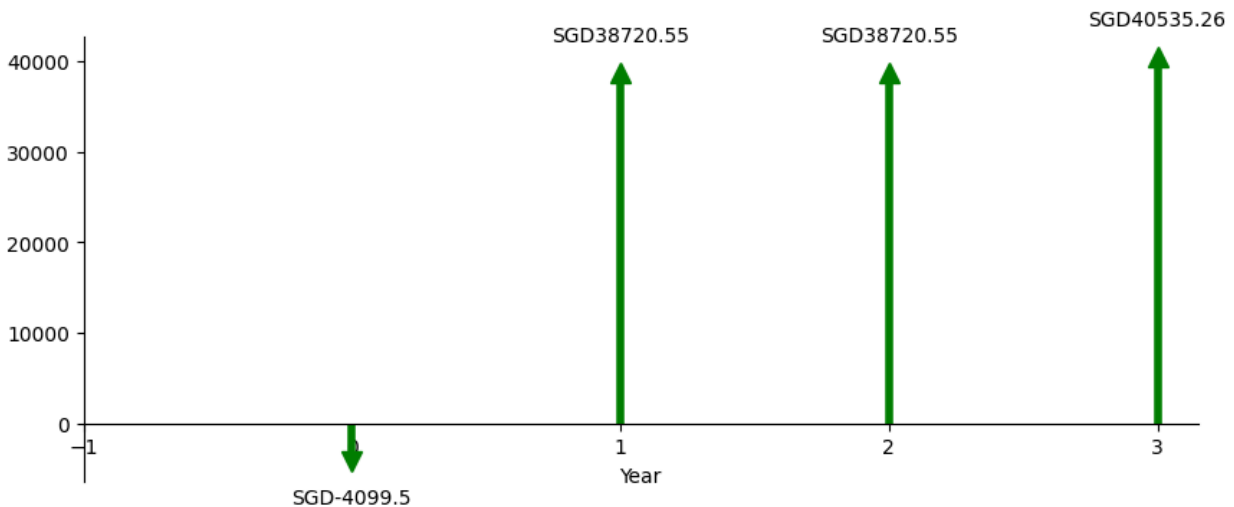
2.1.1. Base Value Data

Factor	Description	Base Data
Initial Investment	One-off cost to purchase machinery (<i>2 Sets of Hardware and Software System</i>)	\$5,466.00
Initial One-off Grant Subsidy	One-off subsidy offered by the Productivity Solutions Grant	\$1,366.50
Annual Sales Volume	Annual retail store revenue	\$1,061,536.65

Annual Benefit	Increase in customer spending owed to customer service provided by cashiers	\$90,230.62
Annual O&M Cost	Inclusive of hardware and software upkeep costs (Grubhub, 2023)	\$2,100.00
Annual Manpower Cost	Two full-time retail associate	\$41,600.00
Shrinkage Cost	Shrinkage costs due to human errors during checkout (Johnson, 2023)	\$159.23
Market Value at EoY 3	-	\$2,186.40
Useful Life (years)	-	5

2.1.2. Cash Flow Model

After-Tax Cash Flow of Alternative 1 Base Case



	(a)	(b)	(c) = (a) - (b)	(d) = -t(c)	(e) = (a) + (d)	(e)/(1+i)^k
End of Year	BTCF	Depreciation	Taxable Income	Tax Cashflow	ATCF	PW of ATCF
0	-\$ 4,099.50				-\$ 4,099.50	-\$ 4,099.50
1	\$ 46,371.38	\$ 1,366.50	\$ 45,004.88	-\$ 7,650.83	\$ 38,720.55	\$ 35,200.50
2	\$ 46,371.38	\$ 1,366.50	\$ 45,004.88	-\$ 7,650.83	\$ 38,720.55	\$ 32,000.46
3	\$ 46,371.38	\$ 1,366.50	\$ 45,004.88	-\$ 7,650.83	\$ 38,720.55	\$ 29,091.33
3	\$ 2,186.40		\$ 2,186.40	-\$ 371.69	\$ 1,814.71	\$ 1,363.42

$$\text{Income Tax Cash Flow at EoY 3} = -0.17 \times 2,186.40 \\ = \underline{\$1,363.42}$$

$$\text{PW(10\%)} \text{ of Alternative 1} = -4,099.50 + \frac{38,720.55}{1.1} + \frac{38,720.55}{1.1^2} + \frac{38,720.55 + 1,814.71}{1.1^3} \\ = \underline{\$93,556.21}$$

2.2. Alternative 2 – Four Unit Self-Checkout System

In this alternative, the scenario where customers are responsible for their own purchases' checkout process, by using a four unit self-checkout system (with one staff on the ground to provide support) is evaluated.

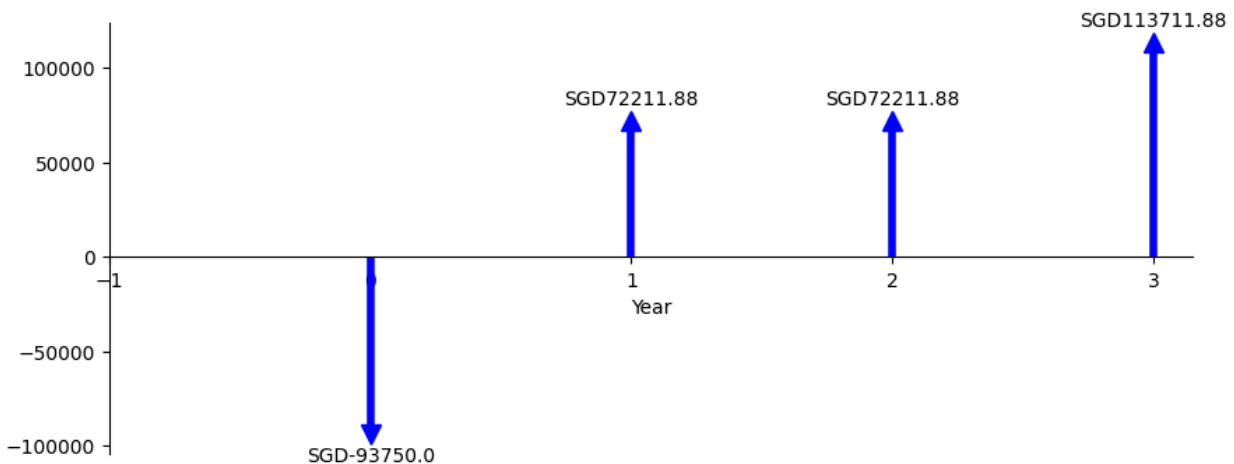
2.2.1. Base Value Data

Factor	Description	Base Data
Initial Investment	One-off cost to purchase machinery (<i>4 Sets of Hardware and Software System</i>)	\$125,000.00
Initial One-off Grant Subsidy	One-off subsidy offered by the Productivity Solutions Grant	\$31,250.00
Annual Sales Volume	Annual retail store revenue	\$1,061,536.65
Annual Benefit	Increase in customer spending due to usage of self-service kiosks	\$159,230.50
Annual O&M Cost	Inclusive of hardware, software upkeep, and vendor costs	\$20,674.05
Annual Manpower Cost	One full-time retail associate	\$20,800.00
Shrinkage Cost	Shrinkage due to higher recorded theft at self-checkout kiosks	\$37,153.78
Market Value at EoY 3	Resale value of the machinery at the end of 3 years	\$50,000.00

Useful Life (years)	The duration that the machinery can operate	5
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2.2.2. Cash Flow Model

After-Tax Cash Flow of Alternative 2 Base Case



	(a)	(b)	(c) = (a) - (b)	(d) = -t(c)	(e) = (a) + (d)	(e)/(1+i)^k
End of Year	BTCF	Depreciation	Taxable Income	Tax Cashflow	ATCF	PW of ATCF
0	-\$ 93,750.00				-\$ 93,750.00	-\$ 93,750.00
1	\$ 80,601.67	\$ 31,250.00	\$ 49,351.67	-\$ 8,389.78	\$ 72,211.88	\$ 65,647.17
2	\$ 80,601.67	\$ 31,250.00	\$ 49,351.67	-\$ 8,389.78	\$ 72,211.88	\$ 59,679.24
3	\$ 80,601.67	\$ 31,250.00	\$ 49,351.67	-\$ 8,389.78	\$ 72,211.88	\$ 54,253.86
3	\$ 50,000.00		\$ 50,000.00	-\$ 8,500.00	\$ 41,500.00	\$ 31,179.56

$$\text{Income Tax Cash Flow at EoY 3} = (50,000 - 0) \times 0.17$$

$$= \underline{\$8,500}$$

$$\text{PW}(10\%) \text{ of Alternative 2} = -93,750 + \frac{72,211.88}{1.1} + \frac{72,211.88}{1.1^2} + \frac{72,211.88 + 41,500}{1.1^3}$$

$$= \underline{\$117,009.83}$$

2.3. Alternative 3 – Checkout-free / Unmanned System

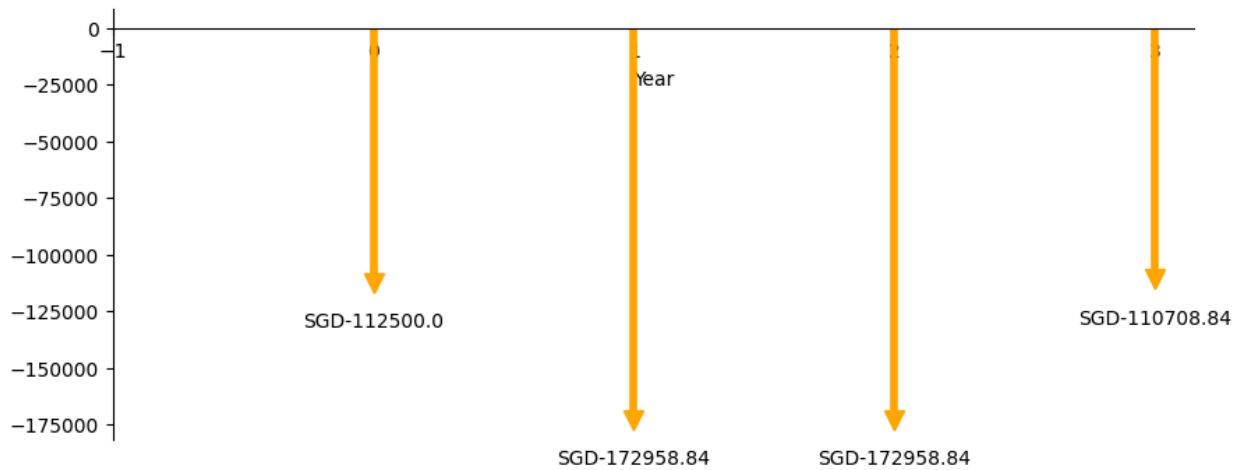
In this alternative, the scenario where a retail outlet is unmanned and fully automated by a Pick 'N' Go system is evaluated.

2.3.1. Base Value Data

Factor	Description	Base Data
Initial Investment	One-off cost to purchase machinery (<i>1 Set of Hardware and Software System</i>)	\$150,000.00
Initial One-off Grant Subsidy	One-off subsidy offered by the Productivity Solutions Grant	\$37,500.00
Annual Sales Volume	Annual retail store revenue	\$1,061,536.65
Annual Benefit	Increase in customer spending due to longer operating hours and convenience	\$53,076.83
Annual O&M Cost	Inclusive of hardware and software upkeep costs	\$250,694.02
Annual Manpower Cost	One part-time retail associate for occasional restocking and customer support	\$13,140.00
Shrinkage Cost	Shrinkage due to failures by the system to recognise products taken out from shelves	\$5,307.68
Market Value at EoY 3	Resale value of the entire system at the end of 3 years	\$75,000.00
Useful Life (years)	The duration that the system can operate	4

2.3.2. Cash Flow Model

After-Tax Cash Flow of Alternative 3 Base Case



	(a)	(b)	(c) = (a) - (b)	(d) = -t(c)	(e) = (a) + (d)	(e)/(1+i)^k
End of Year	BTCF	Depreciation	Taxable Income	Tax Cashflow	ATCF	PW of ATCF
0	-\$ 112,500.00				-\$ 112,500.00	-\$ 112,500.00
1	-\$ 216,064.87	\$ 37,500.00	-\$ 253,564.87	\$ 43,106.03	-\$ 172,958.84	-\$ 157,235.31
2	-\$ 216,064.87	\$ 37,500.00	-\$ 253,564.87	\$ 43,106.03	-\$ 172,958.84	-\$ 142,941.19
3	-\$ 216,064.87	\$ 37,500.00	-\$ 253,564.87	\$ 43,106.03	-\$ 172,958.84	-\$ 129,946.54
3	\$ 75,000.00		\$ 75,000.00	-\$ 12,750.00	\$ 62,250.00	\$ 46,769.35

$$\text{Income Tax Cash Flow at EoY 3} = (75,000 - 0) \times 0.17$$

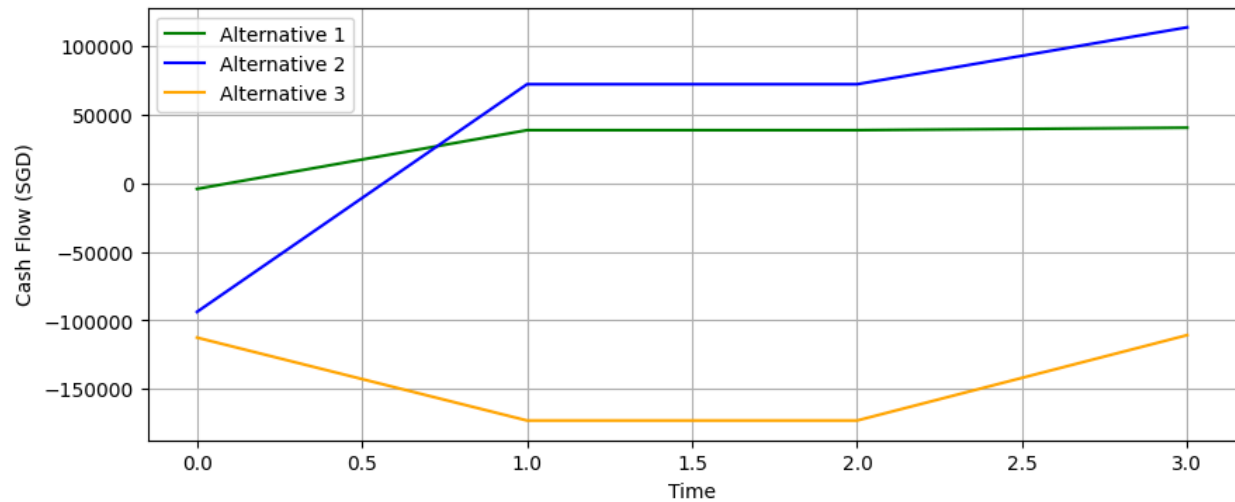
$$= \underline{\$12,750}$$

$$\text{PW}(10\%) \text{ of Alt 3} = -112,500 + \frac{-172,958.84}{1.1} + \frac{-172,958.84}{1.1^2} + \frac{-172,958.84 + 62,250}{1.1^3}$$

$$= \underline{\underline{-\$495,853.69}}$$

2.4. Best Alternative and Solutions (Base-Value Data)

After-Tax Cash Flows of Different Alternatives



PW of Alternative 1 (Conventional Checkout System)	PW of Alternative 2 (Self-checkout System)	PW of Alternative 3 (Checkout-free / Unmanned System)
\$93,556.21	\$117,009.83	-\$495,853.69

From the base cases, Alternative 2 offers the highest PW, and thus is chosen.

The PW of Alternatives 1 and 2 are positive, and are thus feasible projects.

The PW of Alternative 3 is negative, making it an infeasible option to invest in, considering the study period of 3 years. This can be attributed to the highest initial investment cost of Alternative 3, which requires high-end technology and complementary physical products, such as product tags.

3. KEY UNCERTAINTIES

3.1. Sensitivity of Factors (One-Way Range Sensitivity)

To analyse sensitivity factors of each alternative, we generated tornado charts with the help of Excel and Sensit.

Alternative 1 - Conventional Checkout System

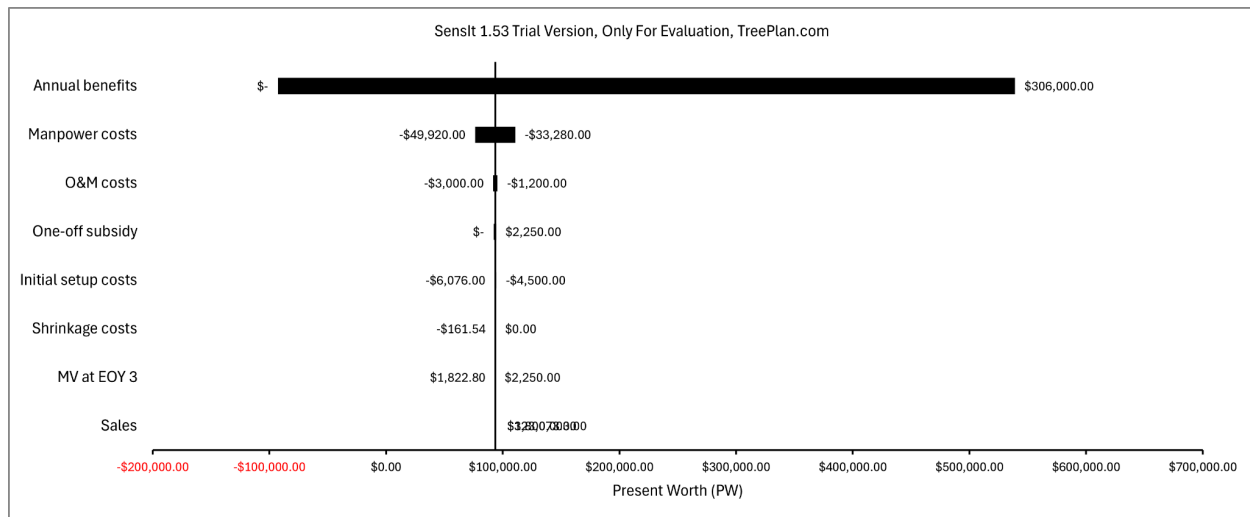


Fig 1. Tornado Chart (Alternative I)

From the tornado chart, it can be analysed that the three most sensitive factors include the Annual Benefits, Manpower Cost, and Operations & Maintenance (O&M) Cost.

As the initial setup costs, market value and shrinkage costs for Alternative 1 are relatively low in comparison with the magnitude of other costs and benefits, it is logical for these factors to be significant in affecting the PW of Alternative 1, as this alternative relies a lot on manpower, so a focus on these factors is not as impactfully substantial as others.

Alternative 2 - Self-checkout System

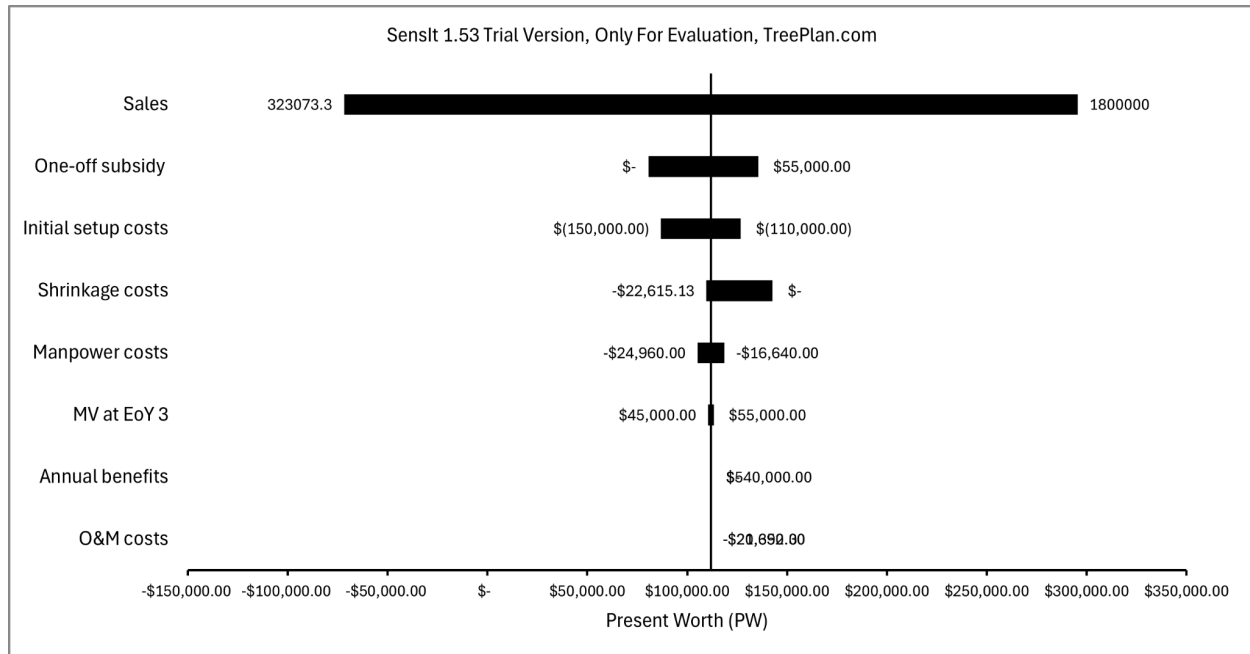


Fig 2. Tornado Chart (Alternative 2)

From the Alternative 2 Tornado Chart, it can be analysed that the Annual Sales Volume, One-off Subsidy, and Initial Investment cost are the three most sensitive factors.

When having an integration of self-checkout and a manned checkout, it is shown that it is mostly reliant on its sales. There is also a less sensitive consideration of Operations & Maintenance cost, and to analyse this may be because the availability of the manpower is still useful making it not as sensitive as other factors. Interestingly, subsidy is the most sensitive here, this may be because it is a combination of both manpower and machinery making it more impactful since it is two aspects of subsidy one is dealing with.

Alternative 3 - Checkout-free / Unmanned System

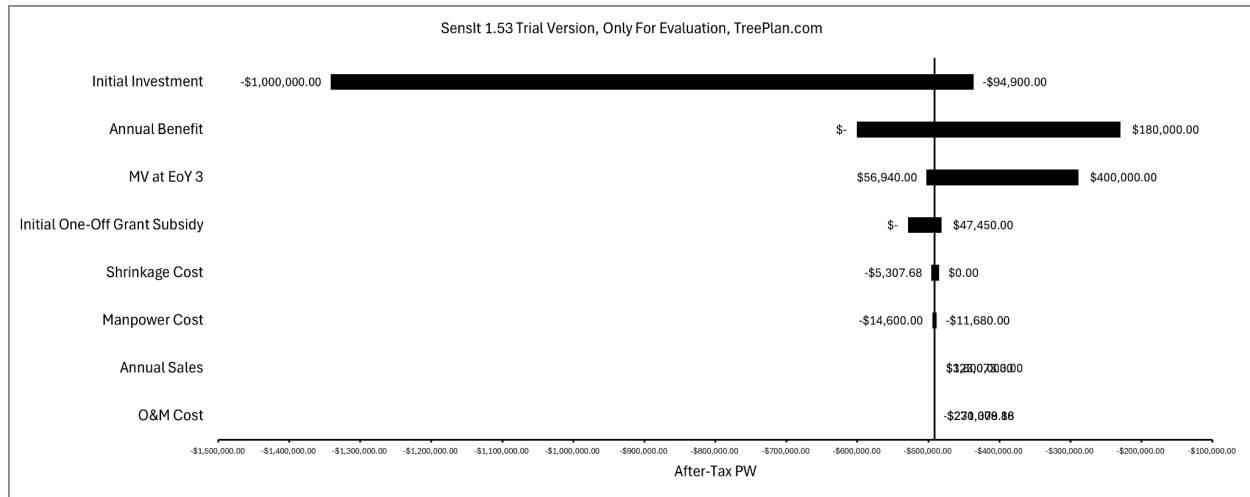


Fig 3. Tornado Chart (Alternative 3)

Lastly, for Alternative 3, it is analysed that the three most sensitive factors include Initial Investment, Annual Benefits, and Market Value at the EoY 3.

In this alternative, manpower costs are pushed down to the bottom. Adding on to that, Annual Sales and Operation & Maintenance costs are the least sensitive compared to other factors, which is the opposite to Alternative 1. This is logical since Alternative 3 deals with more machinery and relies on it, which shows how important the initial investment is and the choice, which also affects the PW itself.

4. PROBABILISTIC RISK ANALYSIS

To analyse probabilistic risk, two models were simulated with differing distributions using both Python and @Risk.

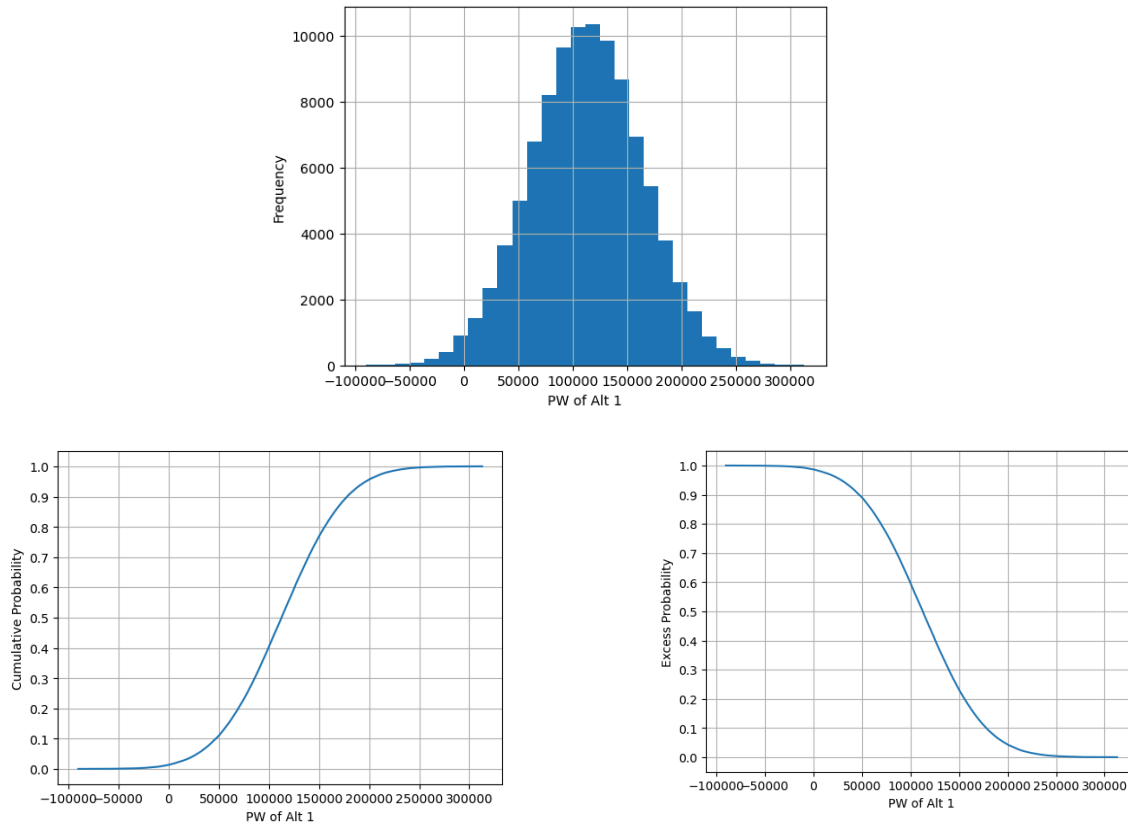
4.1. Simulation I with Python

In the first simulation with Python, the following distributions for the various costs and benefits were selected.

Initial Investment, I	Triangular	A range of possible values are available and the most likely estimate is known
Initial One-off Grant Subsidy, G	Triangular	
Annual Benefits, B	Normal	Can be influenced by many small independent factors affecting shoppers
Annual O&M Costs, E	Triangular	A range of possible values are available and the most likely estimate is known
Annual Manpower Costs, M	Uniform	A range of possible salaries or wages is available, with each value within that range being equally probable based on the qualifications of applicants
Annual Shrinkage Costs, S	Normal	Affected by various random factors such as theft, error and damage
Market Value at EoY 3, MV	Normal	Affected by market forces and independent various random factors

4.1.1. Alternative 1 - Conventional Checkout System

The distribution of the components of this simulation is found in Appendix A.



Probabilistic Analysis on PW of Alt 1

EV = 112,668.39

SD = 50,926.40

Downside Risks:

Pr(PW of Alt 1 $\leq$ -10,000) = 0.73%

Pr(PW of Alt 1 $\leq$ -5,000) = 1.01%

Pr(PW of Alt 1 $\leq$ 0) = 1.32%

Upside Potentials:

Pr(PW of Alt 1 $\geq$ 50,000) = 89.08%

Pr(PW of Alt 1 $\geq$ 100,000) = 59.78%

Pr(PW of Alt 1 $\geq$ 150,000) = 23.18%

Pr(PW of Alt 1 $\geq$ 200,000) = 4.34%

Pr(PW of Alt 1 $\geq$ 250,000) = 0.35%

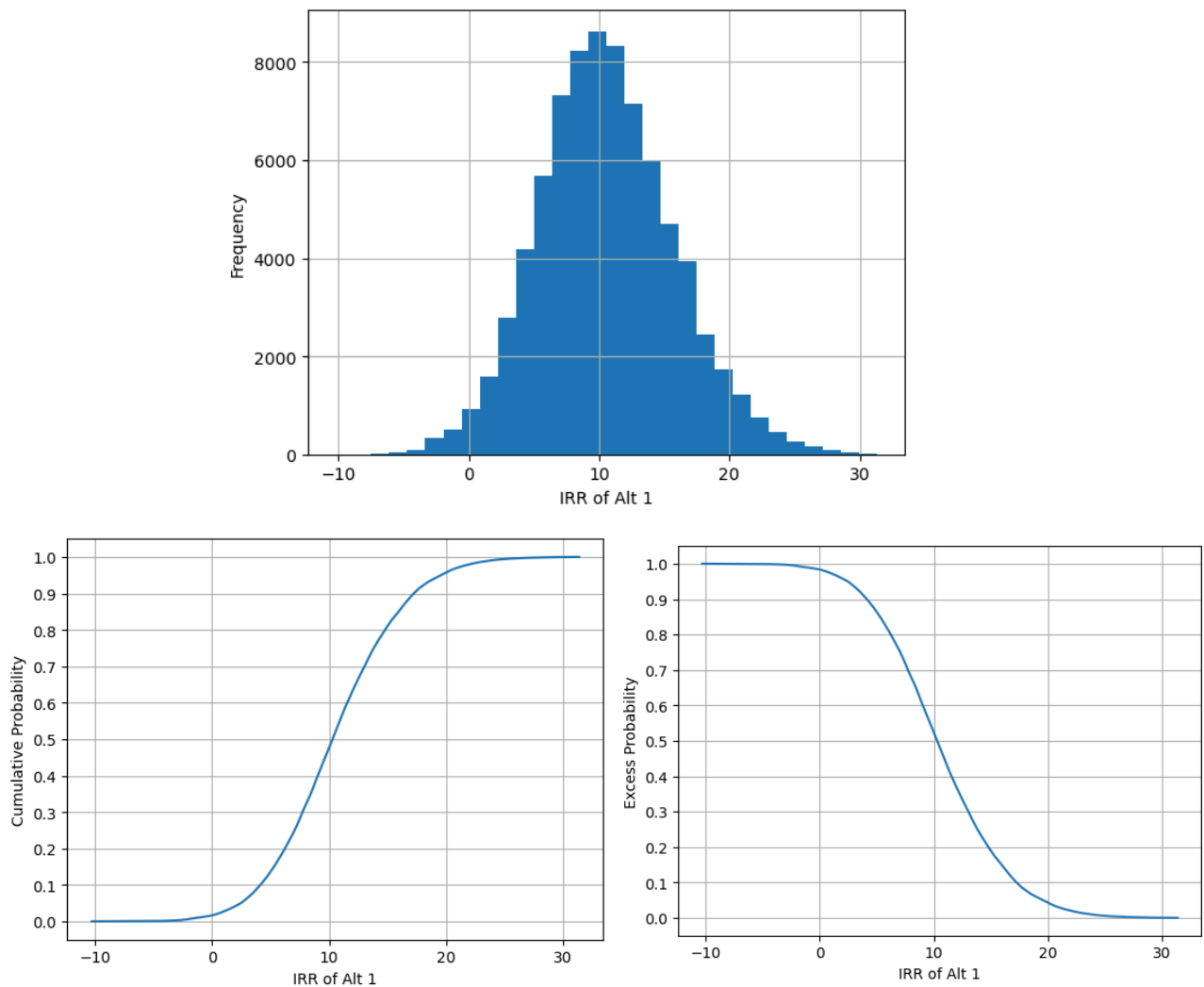
Value-at-Risk:

VaR(99%) = 5,167.38

VaR(95%) = -28,739.08

VaR(90%) = -47,190.01

After 100,000 simulations, Alternative 1 is found to have a mean PW of 112,668.39 and a standard deviation (S.D.) of \$50,926.40. It is a rather financially feasible option with strong probability of positive present worth and a maximum of 1.32% risk of a zero or negative PW.



Probabilistic Analysis on IRR of Alt 1:

EV = 1,052.35%

SD = 516.11%

Downside Risks:

Pr(IRR of Alt 1 $\leq$ 10.0%) = 1.71%

Pr(IRR of Alt 1 $\leq$ 7.5%) = 1.70%

Pr(IRR of Alt 1 $\leq$ 5.0%) = 1.67%

Pr(IRR of Alt 1 $\leq$ 0.0%) = 1.63%

Upside Potentials:

Pr(IRR of Alt 1 $\geq$ 10.0%) = 98.29%

Pr(IRR of Alt 1 $\geq$ 15.0%) = 98.25%

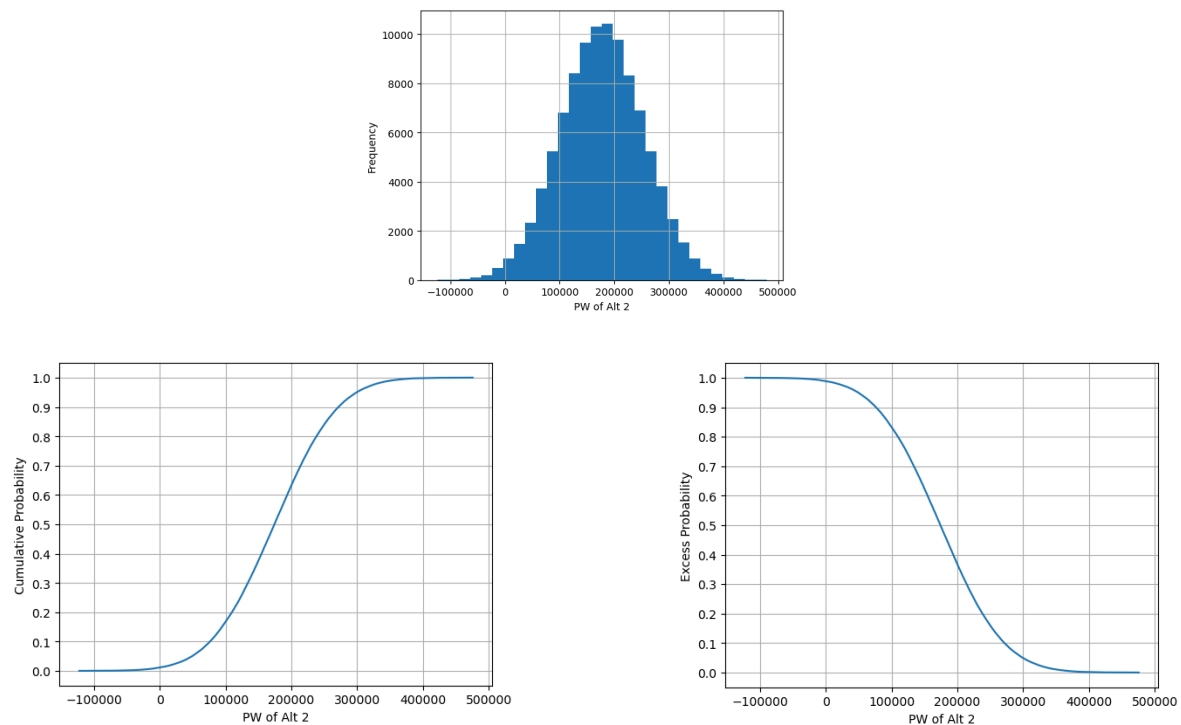
Pr(IRR of Alt 1 $\geq$ 20.0%) = 98.21%

Pr(IRR of Alt 1 $\geq$ 25.0%) = 98.16%

Given the MARR of 10%, investing in Alternative 1 will provide a 98.29% IRR above the MARR of 10%. This makes it a financially feasible and safe option to invest in.

4.1.2. Alternative 2 - Self-checkout System

The distribution of the components of this simulation is found in Appendix B.



Probabilistic Analysis on PW of Alt 2

EV = 177,678.63

SD = 76,102.51

Downside Risks:

Pr(PW of Alt 2 $\leq$ -75,000) = 0.02%

Pr(PW of Alt 2 $\leq$ -50,000) = 0.12%

Pr(PW of Alt 2 $\leq$ -25,000) = 0.33%

Pr(PW of Alt 2 $\leq$ 0) = 0.96%

Upside Potentials:

Pr(PW of Alt 2 $\geq$ 100,000) = 84.50%

Pr(PW of Alt 2 $\geq$ 200,000) = 38.42%

Pr(PW of Alt 2 $\geq$ 300,000) = 5.44%

Pr(PW of Alt 2 $\geq$ 400,000) = 0.16%

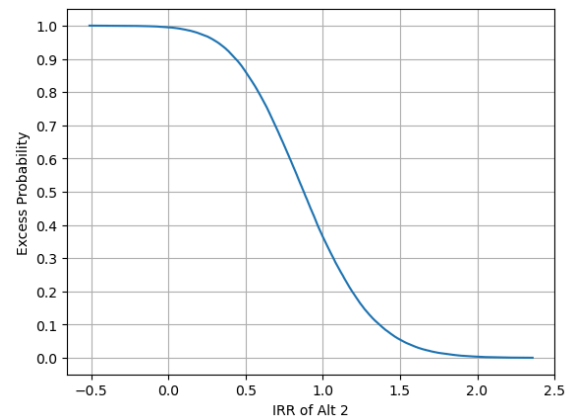
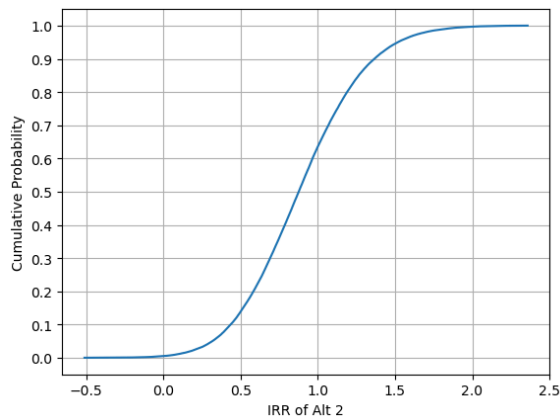
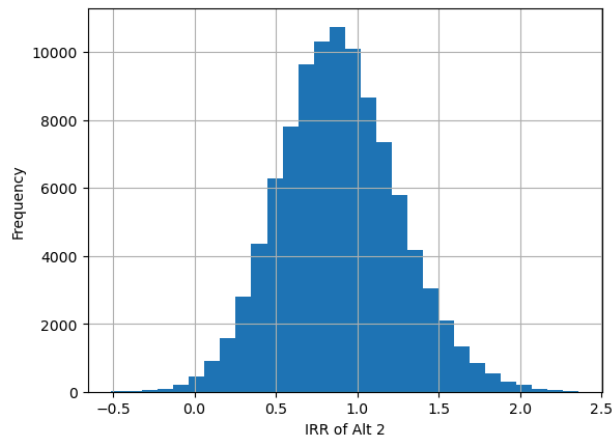
Value-at-Risk:

VaR(99%) = -1,096.74

VaR(95%) = -52,806.13

VaR(90%) = -79,749.07

After 100,000 simulations, Alternative 2 is found to have a mean of 177678.63 and S.D. of 76102.51. Although it has a higher PW than Alternative 1, it does not exhibit mean-variance dominance over Alternative 1. Interestingly, the probability of PW(Alt 2) being a zero or negative value (0.96%) is slightly lower than that of PW(Alt 1) (1.32%). This alternative may appeal to risk-averse retailers who prioritise avoiding losses over the larger variance of PW.



Probabilistic Analysis on IRR of Alt 2:

EV = 90.52%

SD = 36.50%

Downside Risks:

Pr(IRR of Alt 2 ≤ 10.0%) = 0.97%

Pr(IRR of Alt 2 ≤ 7.5%) = 0.79%

Pr(IRR of Alt 2 ≤ 5.0%) = 0.62%

Pr(IRR of Alt 2 ≤ 0.0%) = 0.39%

Upside Potentials:

Pr(IRR of Alt 2 ≥ 10.0%) = 99.03%

Pr(IRR of Alt 2 ≥ 15.0%) = 98.56%

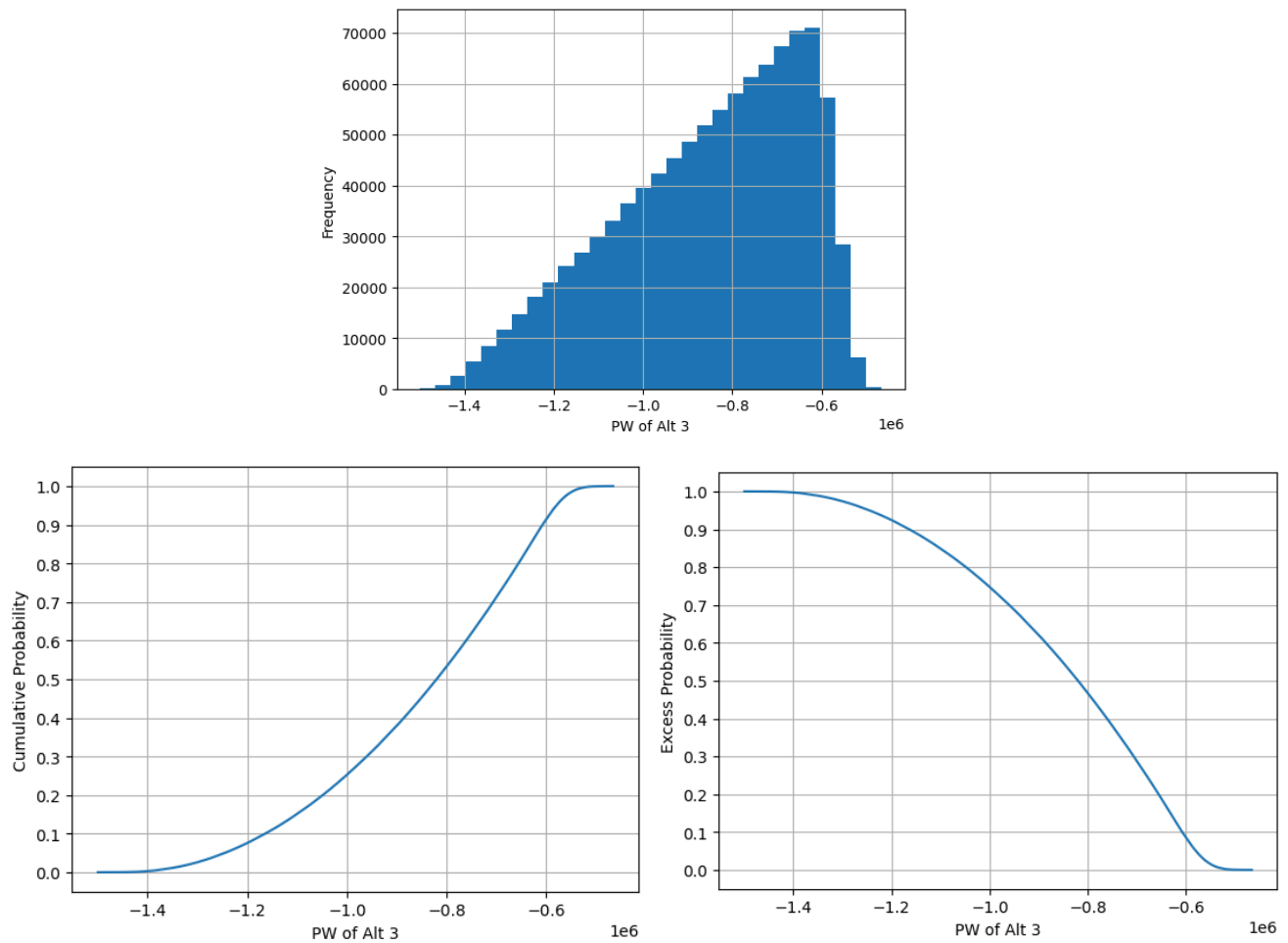
Pr(IRR of Alt 2 ≥ 20.0%) = 97.92%

Pr(IRR of Alt 2 ≥ 25.0%) = 97.06%

Similarly, the probability that Alternative 2 generates a higher IRR than the MARR (99.03%) is slightly higher than that of Alternative 1 (98.29%). Nonetheless, to make a distinction between the 2 alternatives, retailers have to firstly consider the upfront costs they are willing or able to pay as it is significantly higher for Alternative 2.

4.1.3. Alternative 3 - Checkout-free / Unmanned System

The distribution of the components of this simulation is found in Appendix C.



Probabilistic Analysis on PW of Alt 3

EV = -854,891.27

SD = 208,592.94

Downside Risks:

$\Pr(\text{PW of Alt 3} \leq -800,000) = 53.31\%$

$\Pr(\text{PW of Alt 3} \leq -700,000) = 71.25\%$

$\Pr(\text{PW of Alt 3} \leq -600,000) = 91.49\%$

$\Pr(\text{PW of Alt 3} \leq -500,000) = 99.96\%$

Upside Potentials:

$\Pr(\text{PW of Alt 3} \geq 100,000) = 0.00\%$

Value-at-Risk:

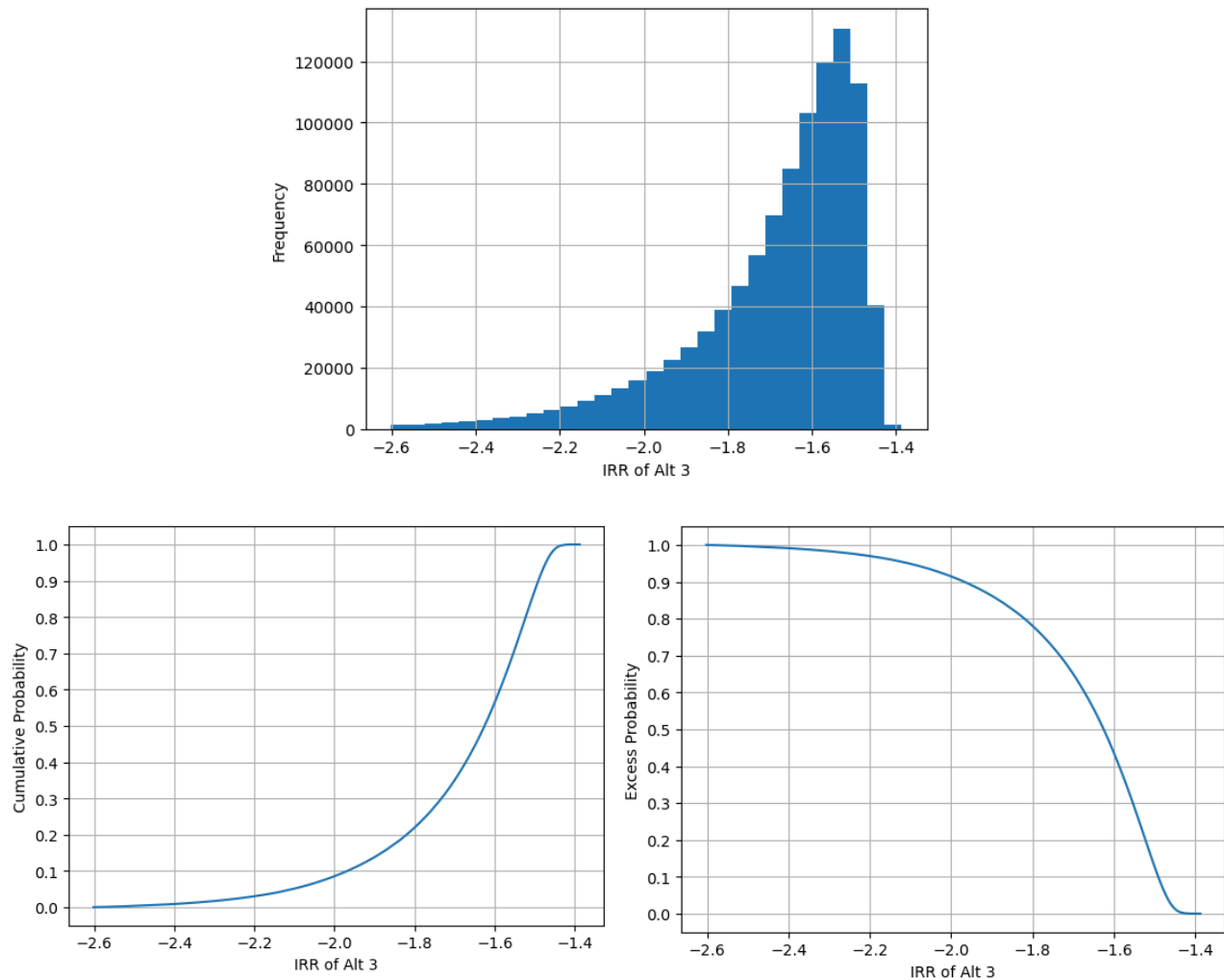
VaR(99%) = 1,355,783.61

VaR(95%) = 1,245,039.66

VaR(90%) = 1,163,984.79

After 1,000,000 simulations, PW(Alternative 3) is still left-skewed due to the significantly left-skewed triangular distribution that is large in magnitude relative to other cash flows.

Regardless, Alternative 3 offers no upside potential and significant downside risks. This could be accounted for by the high initial investment and operational costs and lower benefits provided according to research. The study period of 3 years makes it challenging for these costs to be spread out sufficiently to justify its investment.



Probabilistic Analysis on IRR of Alt 3:

EV = -168.06%

SD = 19.53%

Downside Risks:

Pr(IRR of Alt 3 ≤ 10.0%) = 100.00%

Pr(IRR of Alt 3 ≤ 5.0%) = 100.00%

Pr(IRR of Alt 3 ≤ 0.0%) = 100.00%

Pr(IRR of Alt 3 ≤ -40.0%) = 100.00%

Upside Potentials:

Pr(IRR of Alt 3 ≥ 1.0%) = 0.00%

Pr(IRR of Alt 3 ≥ 15.0%) = 0.00%

Pr(IRR of Alt 3 ≥ 20.0%) = 0.00%

Pr(IRR of Alt 3 ≥ 25.0%) = 0.00%

Since Alternative 3 does not generate a positive IRR within a study period of 3 years, it is not financially feasible for retailers to select this option.

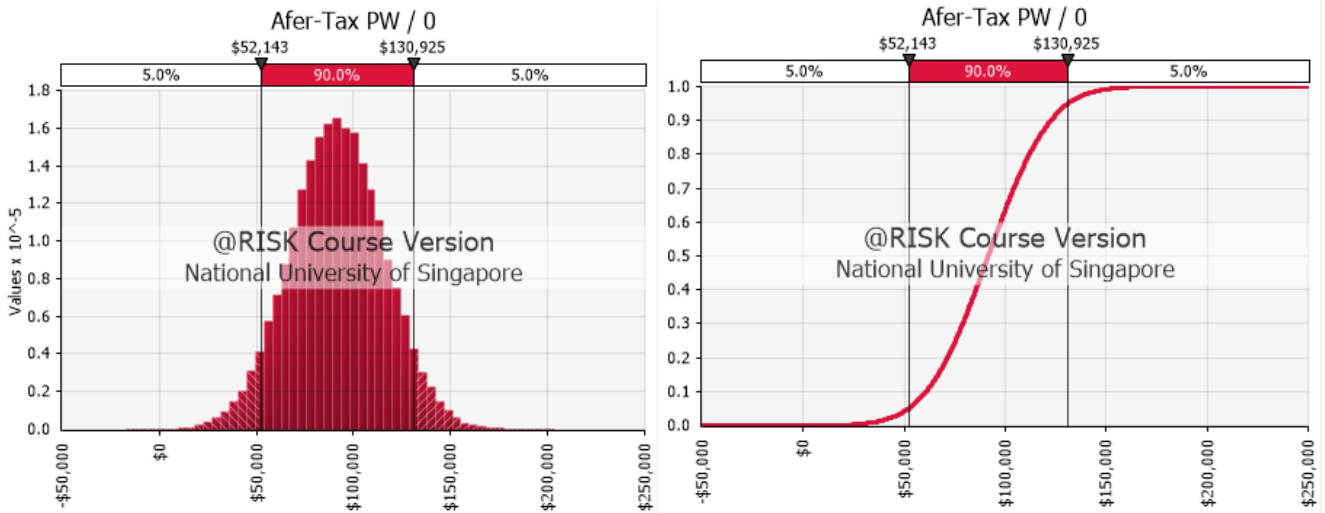
Both Alternatives 1 and 2 exhibit mean-variance dominance over Alternative 3.

To conclude, with a study period of 3 years and MARR of 10%, Alternatives 1 and 2 offer relatively similar positive returns and probabilities of upside and downside risks. As such, based on this simulation and probabilistic risk analysis, retailers can reap the benefits of either alternative, but should make a decision based on the upfront costs they are willing to invest for a checkout system.

4.2. Simulation II with @Risk

Initial Investment, I	Triangular	A range of possible values are available and the most likely estimate is known
Initial One-off Grant Subsidy, G	Triangular	
Annual Benefits, B	Normal	Can be influenced by many small independent factors affecting shoppers
Annual O&M Costs, E	Lognormal	Account for the multiplicative effects of changes in operating costs
Annual Manpower Costs, M	Lognormal	Account for the multiplicative effects of changes in manpower costs
Annual Shrinkage Costs, S	Triangular	A range of possible values are available and the most likely estimate is known
Market Value at EoY 3, MV	Normal	Affected by market forces and independent various random factors

4.2.1 Alternative 1 - Conventional Checkout System



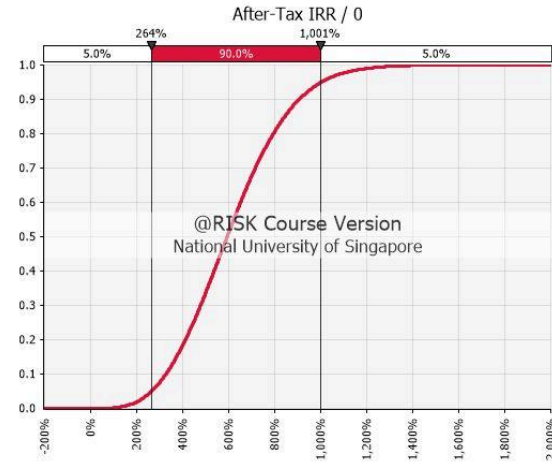
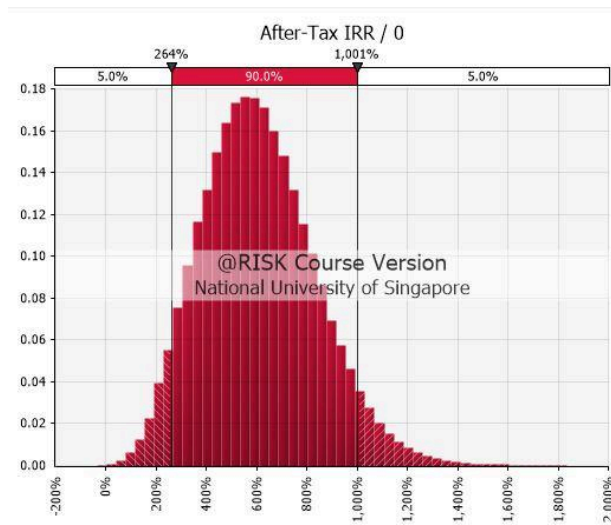
Expected After-Tax PW	\$91,556.58
Variance of After-Tax PW	\$578,772,929.30
Standard Deviation of After-Tax PW	\$24,057.70
Upside-Downside Potential	Downside Risks: ● $P(\text{PW of Alt 1} \leq 15,000) = 0.1\%$ ● $P(\text{PW of Alt 1} \leq 25,000) = 0.3\%$ ● $P(\text{PW of Alt 1} \leq 35,000) = 0.9\%$ Upside Potentials: ● $P(\text{PW of Alt 1} \geq 70,000) = 81.5\%$ ● $P(\text{PW of Alt 1} \geq 100,000) = 36.4\%$ ● $P(\text{PW of Alt 1} \geq 130,000) = 5.4\%$ ● $P(\text{PW of Alt 1} \geq 160,000) = 0.2\%$

Through the simulation in @RISK, it can be found that Alternative 1 has an extremely large variance of \$578,772,929.30 and a standard deviation of \$24,057.70, suggesting that the after-tax PW of a conventional checkout system is subjected to significant variability.

Referring back to the tornado chart in *Part 3.1*, it can be deduced that this variability depends highly on the Annual Benefit, Manpower Cost, as well as the O&M Cost. This relates back to the

annual sales volume as these factors are directly dependent on it; uncertain revenue streams due to changes in market conditions can affect these factors greatly. A recent example of this is the emergence of COVID-19, which has caused quarantines and caused the retail workforce to face a steep decline.

The downside risks meanwhile show that there is a minimal chance of a conventional checkout system underperforming. This implies that the system is relatively stable and less prone to extreme losses, factually proven by the long success and existence of the system since 1916 (Ross, 2016) . The upside potentials, meanwhile, shows high probabilities of achieving positive returns. However, as the thresholds increase, the probabilities decrease significantly, indicating much lower returns at higher levels.



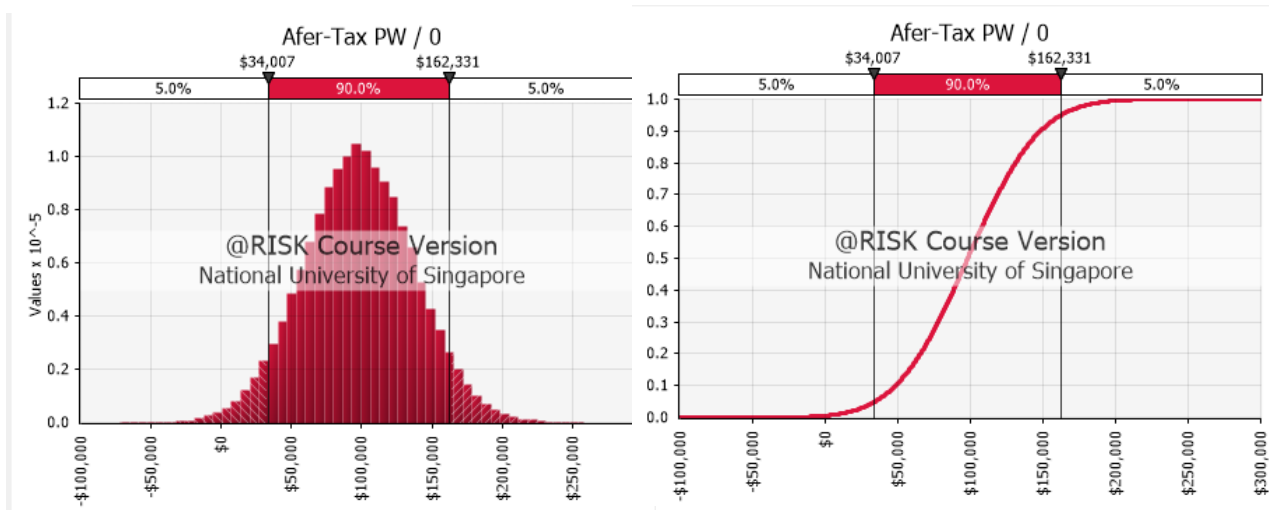
Expected After-Tax IRR	607.4%
Variance of After-Tax IRR	50,805.16%
Standard Deviation of After-Tax IRR	225.4%
Upside-Downside Potential	Downside Risks: ● P(IRR of Alt 1 <= 200%) = 1.9% ● P(IRR of Alt 1 <= 150%) = 0.8% ● P(IRR of Alt 1 <= 100%) = 0.2% Upside Potentials: ● P(IRR of Alt 1 >= 100%) = 99.8%

- $P(\text{IRR of Alt 1} \geq 150\%) = 99.2\%$
- $P(\text{IRR of Alt 1} \geq 200\%) = 98.1\%$
- $P(\text{IRR of Alt 1} \geq 250\%) = 95.8\%$

For the IRR, its variance is quite large being 50,8505.16%, suggesting substantial variability in the monetary returns of the checkout system. Its standard deviation also indicates the relatively high volatility of the returns.

Its downside risks, meanwhile, are relatively low. For example, the probability of the IRR being less than 200% is 1.9%, showing a small chance of low returns. Conversely, the probabilities of achieving high IRRs are extremely high. Hence, while there are some risks of lower-than-expected returns, the probabilities of achieving high returns far outweigh the probabilities of experiencing significant loss.

4.2.2 Alternative 2 - Self-checkout System



Expected After-Tax PW	\$98,167.52
Variance of After-Tax PW	\$1,514,292,391.00
Standard Deviation of After-Tax PW	\$38,913.91
	Downside Risks: • $P(\text{PW of Alt 2} \leq -10,000) = 0.3\%$ • $P(\text{PW of Alt 2} \leq 0) = 0.6\%$

Upside-Downside Potential

- $P(\text{PW of Alt 2} \leq 10,000) = 1.1\%$

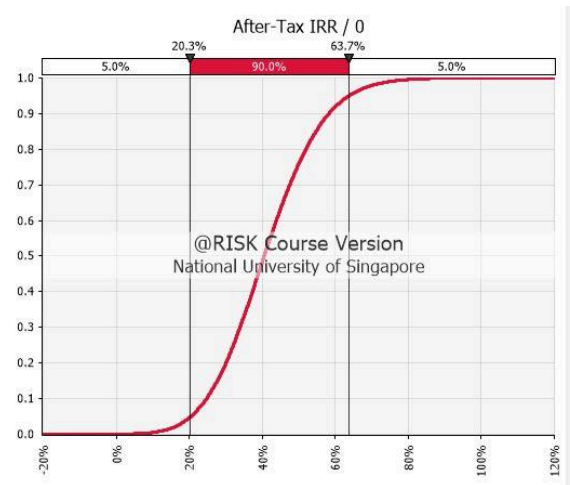
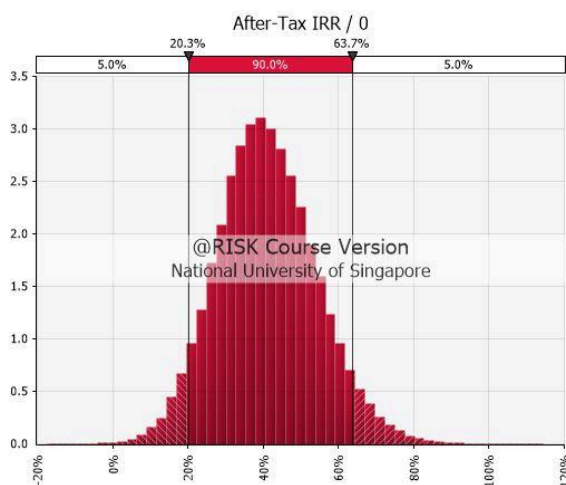
Upside Potentials:

- $P(\text{PW of Alt 2} \geq 60,000) = 83.7\%$
- $P(\text{PW of Alt 2} \geq 110,000) = 37.9\%$
- $P(\text{PW of Alt 2} \geq 160,000) = 5.6\%$
- $P(\text{PW of Alt 2} \geq 210,000) = 0.2\%$

For Alternative 2, it can be found that it has a much higher variance and standard deviation value as compared to Alternative 1. This suggests that the self-checkout system has a more volatile PW, which can be due to technological factors associated with the machines.

The downside risks for Alternative 2 are slightly higher as compared to Alternative 1. Despite this, the probabilities remain low, still indicating minimal significant loss. The upside potential, meanwhile, has a similar result as compared to Alternative 1 - probabilities decreasing as the thresholds increase. Interestingly, this finding is different from the result from Simulation 1, which suggests that the different distribution used affected the comparative upside and downside risks, albeit not significantly.

Therefore, while Alternative 2 is expected to yield higher after-tax returns as compared to Alternative 1 based on the expected after-tax PW, it comes with higher risks. Hence, stakeholders should consider these trade-offs when deciding between the two systems and take into account their risk tolerance and objectives of these investments.

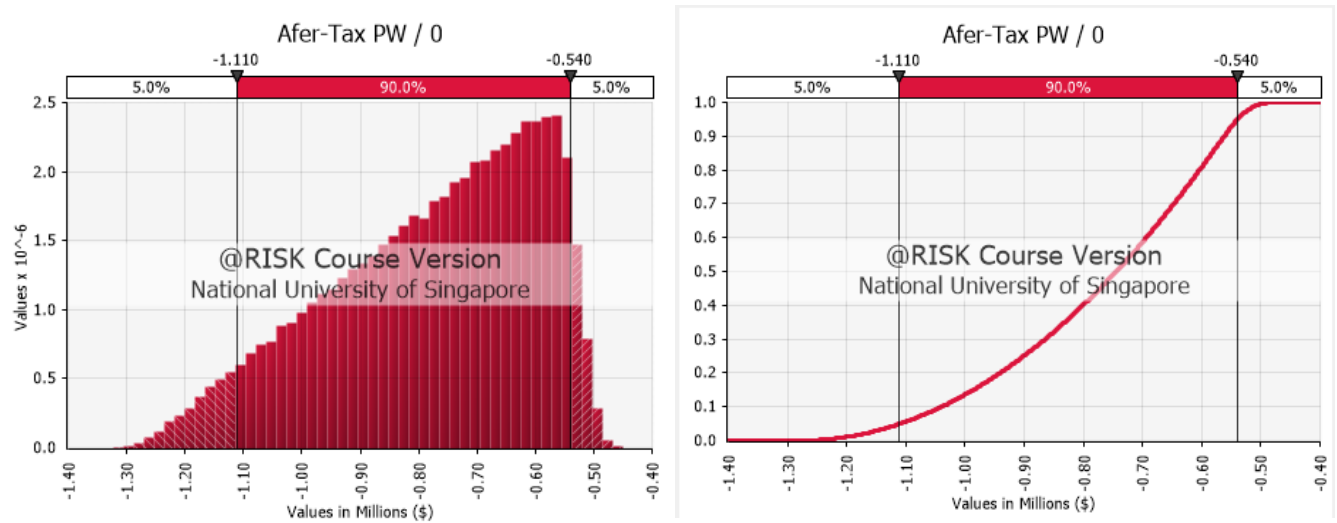


Expected After-Tax IRR	41.09%
Variance of After-Tax IRR	174.24%
Standard Deviation of After-Tax IRR	13.20%
Upside-Downside Potential	Downside Risks: ● $P(\text{IRR of Alt 2} \leq 10\%) = 0.6\%$ ● $P(\text{IRR of Alt 2} \leq 7.5\%) = 0.3\%$ ● $P(\text{IRR of Alt 2} \leq 5\%) = 0.2\%$ ● $P(\text{IRR of Alt 2} \leq 0\%) = 0.0\%$ Upside Potentials: ● $P(\text{IRR of Alt 2} \geq 10\%) = 99.4\%$ ● $P(\text{IRR of Alt 2} \geq 15\%) = 98.3\%$ ● $P(\text{IRR of Alt 2} \geq 20\%) = 95.3\%$ ● $P(\text{IRR of Alt 2} \geq 25\%) = 89.6\%$

For its IRR, Alternative 2 has a lower variance and standard deviation as compared to Alternative 1, indicating lower variability and volatility in profiting from their systems. For downside risks, Alternative 2 has lower probabilities as compared to Alternative 1.

In conclusion, while Alternative 1 has higher returns and greater volatility, Alternative 2 offers a more stable investment option, albeit slightly lower potential returns.

4.2.3 Alternative 3 - Checkout-free / Unmanned System



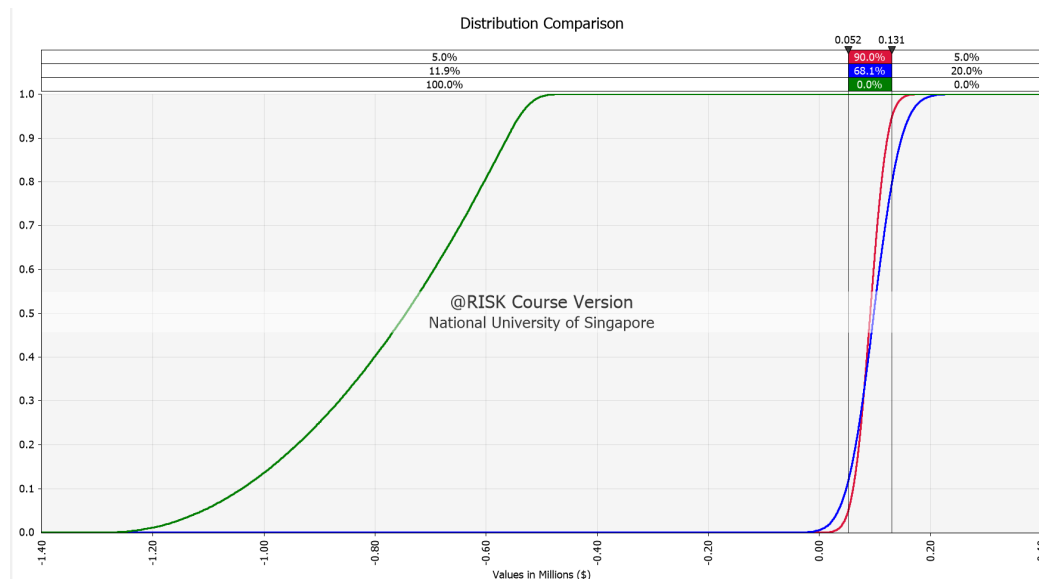
Expected After-Tax PW	-\$774,941.73
Variance of After-Tax PW	\$32,048,360,902.71
Standard Deviation of After-Tax PW	\$179,020.56
Upside-Downside Potential	Downside Risks: • $P(\text{PW of Alt 3} \leq -900,000) = 25.2\%$ • $P(\text{PW of Alt 3} \leq -800,000) = 40.3\%$ • $P(\text{PW of Alt 3} \leq -700,000) = 58.8\%$ • $P(\text{PW of Alt 3} \leq -600,000) = 81.0\%$ • $P(\text{PW of Alt 3} \leq -500,000) = 99.5\%$ Upside Potentials: • $P(\text{PW of Alt 3} \geq 100,000) = 0.0\%$

In Alternative 3, the negative after-tax PW indicates that there will be a significant loss if the checkout-free system is being implemented. In addition to that, it has the highest variance and standard deviation values amongst the three alternatives, indicating that Alternative 3's financial performance is highly unpredictable and subjected to various factors, such as operational challenges.

The downside risks for Alternative 3 are extremely high, with a whopping 99.5 percent chance of a \$500,000 loss. Unlike the previous two alternatives, Alternative 3 has no upside potential, indicating that the system is not able to generate any profit in the three years. Hence, with these results, stakeholders should not be considering implementing the checkout-free system, specifically if its study period is three.

There is no IRR value for Alternative 3 due to all the cash flows being negative. Hence, Alternative 3 is concluded to be infeasible.

4.2.4 Dominance Analysis



Red Line: Alternative 1 - Conventional Checkout System

Blue Line: Alternative 2 - Self-checkout System

Green Line: Alternative 3 - Checkout-free / Unmanned System

Alternative 1 vs. Alternative 2:

There is no stochastic dominance between Alternative 1 and 2. There is also no mean-variance dominance since the expected PW of Alternative 1 is less than that of Alternative 2, but Alternative 2 has a much greater variance than Alternative 1.

Alternative 1 vs. Alternative 3:

Alternative 1 dominates Alternative 3 in both stochastic dominance and mean-variance dominance. Alternative 1 has a much smaller variance than Alternative 3 and has a positive expected after-tax PW, while Alternative 3 has a negative expected after-tax PW.

Alternative 2 vs. Alternative 3:

Alternative 2 dominates Alternative 3 in both stochastic dominance and mean-variance dominance. Similar to the previous comparison of Alternative 1 vs. Alternative 3, Alternative 2's after-tax PW is positive, whereas Alternative 3's expected after-tax PW is negative. Alternative 2 also has a much smaller variance than Alternative 3.

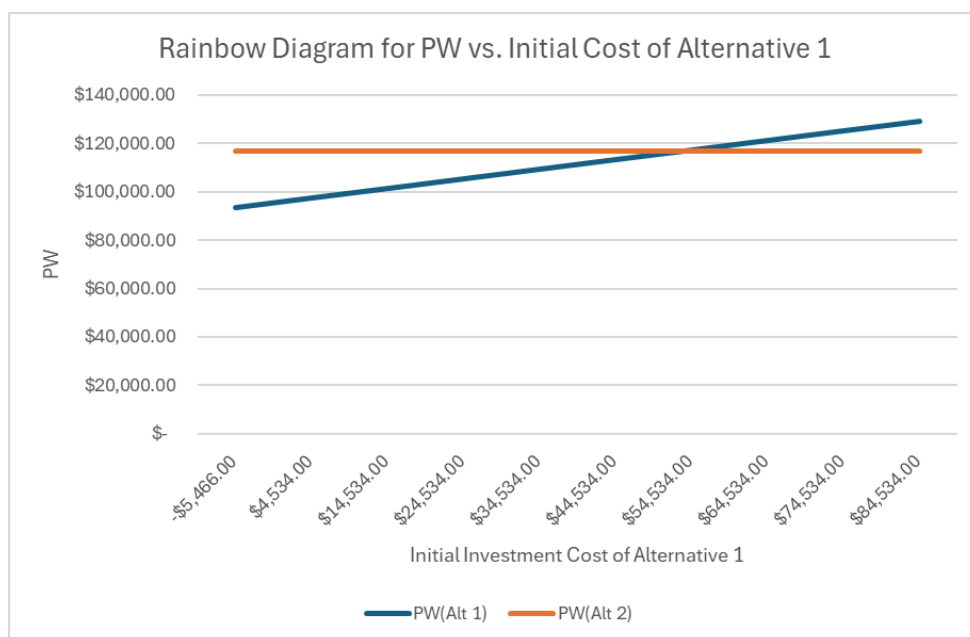
5. WHAT-IF SCENARIO ANALYSIS

Sensitivity Analysis on Investment Cost of Alternative 1 (Comparing Alternatives 1 and 2):

Table 1. Table of PW(Alt 1) against Initial Cost of Alt 1

Initial Cost of Alt 1	PW(Alt 1)	PW(Alt 2)
-\$ 5,466.00	\$ 93,556.21	\$ 117,009.83
\$ 4,534.00	\$ 97,504.93	\$ 117,009.83
\$ 14,534.00	\$ 101,453.65	\$ 117,009.83
\$ 24,534.00	\$ 105,402.38	\$ 117,009.83
\$ 34,534.00	\$ 109,351.10	\$ 117,009.83
\$ 44,534.00	\$ 113,299.82	\$ 117,009.83
\$ 54,534.00	\$ 117,248.54	\$ 117,009.83
\$ 64,534.00	\$ 121,197.27	\$ 117,009.83
\$ 74,534.00	\$ 125,145.99	\$ 117,009.83
\$ 84,534.00	\$ 129,094.71	\$ 117,009.83

Figure 4. Rainbow Diagram of PW vs. Initial Cost of Alternative 1



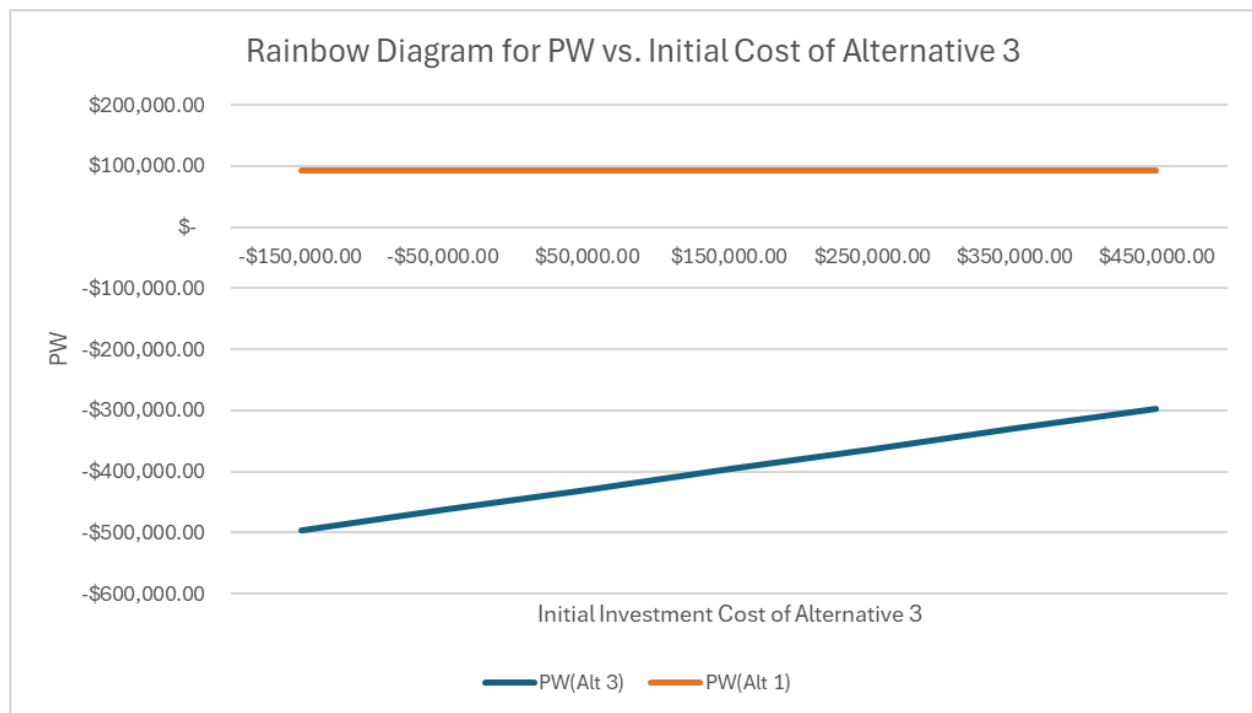
From Table 1, it shows that for $PW(Alt\ 1) > PW(Alt\ 2) = \$117,009.83$, the Initial Cost of Alternative 1 should be positive. This is not possible as the initial cash flow to implement a conventional check-out system is a negative setup cost. Thus, Alternative 2 is always preferred over Alternative 1 for all Initial Costs of Alternative 1.

Sensitivity Analysis on Investment Cost of Alternative 3 (Comparing Alternatives 3 and 1):

Table 2. Table of $PW(Alt\ 3)$ against Initial Cost of Alt 3

Initial Cost of Alt 3	$PW(Alt\ 3)$	$PW(Alt\ 1)$
-\$ 150,000.00	-\$ 495,853.69	\$ 93,556.21
-\$ 50,000.00	-\$ 462,602.38	\$ 93,556.21
\$ 50,000.00	-\$ 429,351.06	\$ 93,556.21
\$ 150,000.00	-\$ 396,099.75	\$ 93,556.21
\$ 250,000.00	-\$ 362,848.43	\$ 93,556.21
\$ 350,000.00	-\$ 329,597.12	\$ 93,556.21
\$ 450,000.00	-\$ 296,345.81	\$ 93,556.21

Figure 5. Rainbow Diagram of PW vs. Initial Cost of Alternative 3



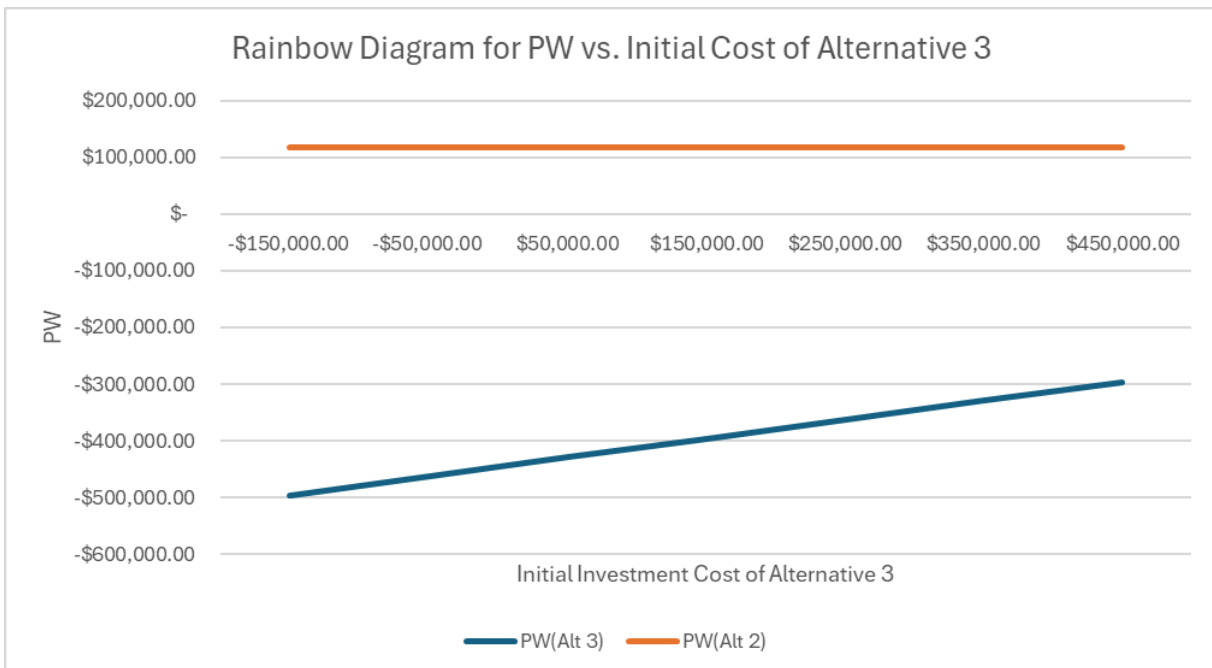
From Table 1, it shows that for $PW(Alt\ 3) > PW(Alt\ 1) = \93556.21 , the Initial Cost of Alternative 3 should be positive. This is not possible as the initial cash flow to implement an unmanned shop is a negative setup cost. Thus, Alternative 1 is always preferred over Alternative 3 for all Initial Costs of Alternative 3.

Sensitivity Analysis on Investment Cost of Alternative 3 (Comparing Alternatives 3 and 2):

Table 3. Table of $PW(Alt\ 3)$ against Initial Cost of Alt 3

Initial Cost of Alt 3	$PW(Alt\ 3)$	$PW(Alt\ 2)$
-\$ 150,000.00	-\$ 495,853.69	\$ 117,009.83
-\$ 50,000.00	-\$ 462,602.38	\$ 117,009.83
\$ 50,000.00	-\$ 429,351.06	\$ 117,009.83
\$ 150,000.00	-\$ 396,099.75	\$ 117,009.83
\$ 250,000.00	-\$ 362,848.43	\$ 117,009.83
\$ 350,000.00	-\$ 329,597.12	\$ 117,009.83
\$ 450,000.00	-\$ 296,345.81	\$ 117,009.83

Figure 6. Rainbow Diagram of PW vs. Initial Cost of Alternative 3



From Table 1, it shows that for $PW(Alt\ 3) > PW(Alt\ 2) = \117009.83 , the Initial Cost of Alternative 3 should be positive. This is not possible as the initial cash flow to implement an

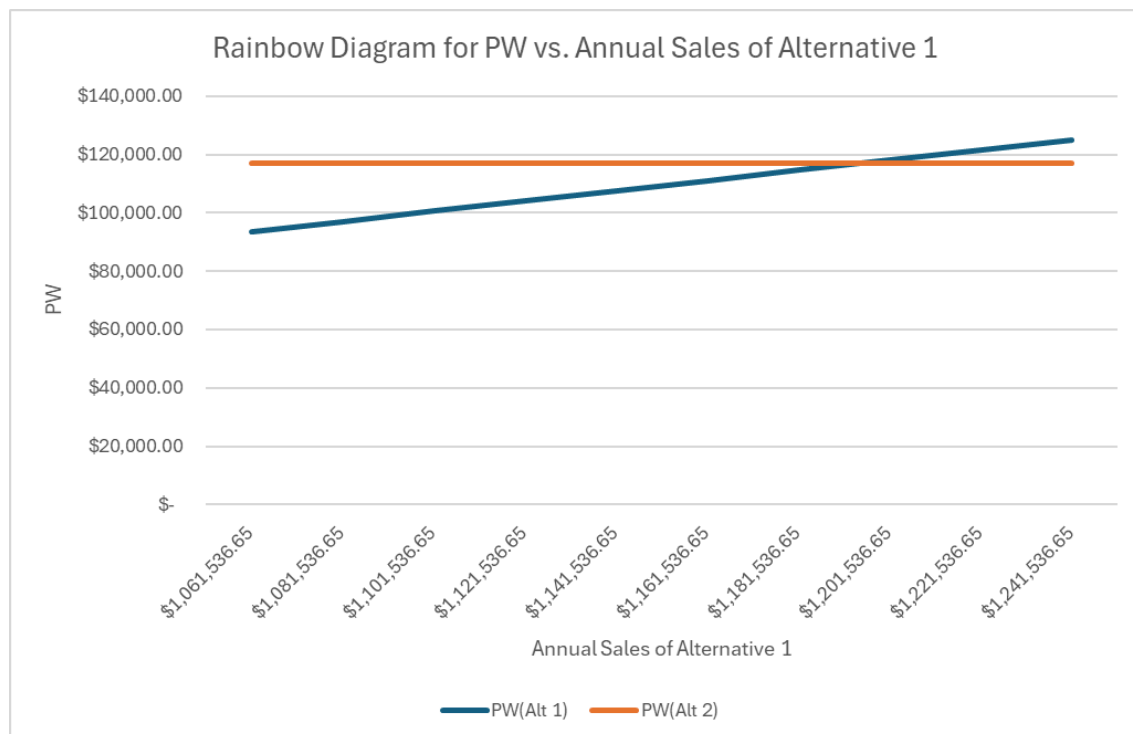
unmanned shop is a negative setup cost. Thus, Alternative 2 is always preferred over Alternative 3 for all Initial Costs of Alternative 3.

Sensitivity Analysis on Annual Sales of Alternative 1 (Comparing Alternatives 1 and 2):

Table 4. Table of PW(Alt 1) against Annual Sales of Alt 1

Annual Sales of Alt 1	PW(Alt 1)	PW(Alt 2)
\$ 1,061,536.65	\$ 93,556.21	\$ 117,009.83
\$ 1,081,536.65	\$ 97,058.96	\$ 117,009.83
\$ 1,101,536.65	\$ 100,561.72	\$ 117,009.83
\$ 1,121,536.65	\$ 104,064.48	\$ 117,009.83
\$ 1,141,536.65	\$ 107,567.23	\$ 117,009.83
\$ 1,161,536.65	\$ 111,069.99	\$ 117,009.83
\$ 1,181,536.65	\$ 114,572.74	\$ 117,009.83
\$ 1,201,536.65	\$ 118,075.50	\$ 117,009.83
\$ 1,221,536.65	\$ 121,578.25	\$ 117,009.83
\$ 1,241,536.65	\$ 125,081.01	\$ 117,009.83

Figure 7. Rainbow Diagram of PW vs. Annual Sales of Alternative 1



Let X = Annual Sales of Alternative 1

Annual Costs of Alternative 1:

= Operation & Maintenance Costs + Manpower Costs + Shrinkage Costs

= - 2100 - 41600 - 159.23

= \$43,859.23

PW of Alternative 1:

$$= -4099.50 + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^1} + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^2} \\ + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^3} + \frac{2186.40-0.17(2186.40)}{(1+0.1)^3}$$

PW of Alternative 2: \$117,009.83

Computing the Break-Even Point:

Solve:

$$-4099.50 + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^1} + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^2} \\ + \frac{(1-0.17)(0.085X-43859.23)+0.17(1366.50)}{(1+0.1)^3} + \frac{2186.40-0.17(2186.40)}{(1+0.1)^3} = \$117,009.83$$

$X = \$1,195,220.00$

Decision Rules based on Annual Sales of Alternative 1:

Annual Sales of Alternative 1:	Optimal Alternative:
$X \leq \$1,195,220$	Alternative 2
$\$1,195,220 \leq X$	Alternative 1

Decision Reversal based on Annual Sales of Alternative 1:

- At base value $X = \$1,061,536.65$, Alternative 2 is preferred to Alternative 1
- Change in value required for a decision reversal
 $= 1195220 - 1061536.65$
 $= \$133,683.35$

- Hence, %-Change required for a decision reversal

$$= \frac{133683.35}{1061536.65}$$

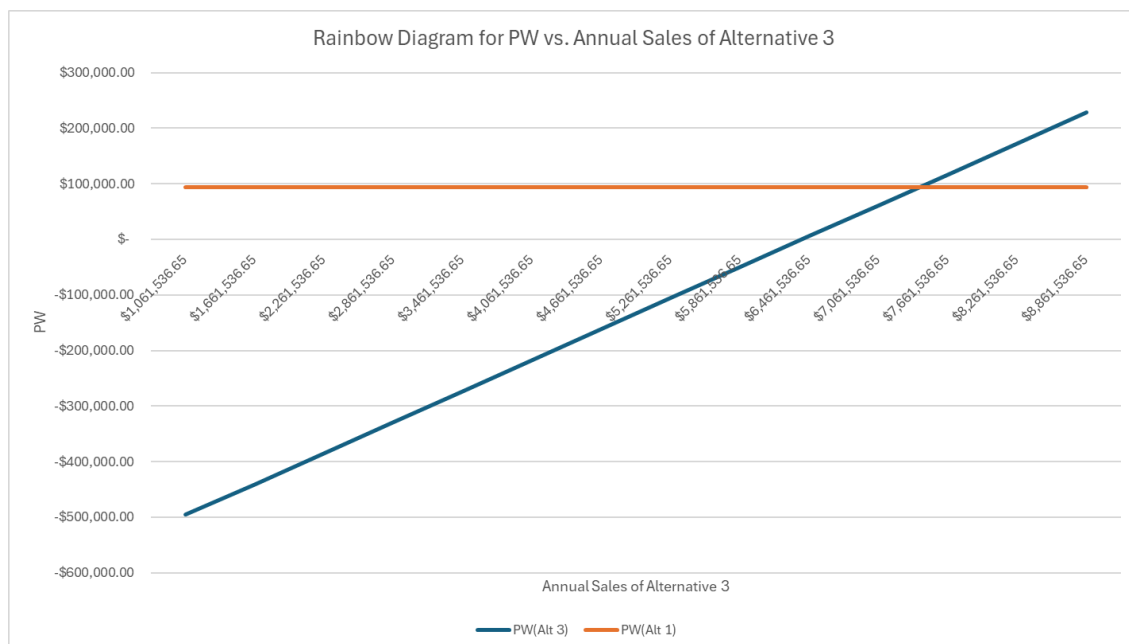
$$= + 12.59\%$$

Sensitivity Analysis on Annual Sales of Alternative 3 (Comparing Alternatives 3 and 1)

Table 5. Table of PW(Alt 3) against Annual Sales of Alt 3

Annual Sales of Alt 3	PW(Alt 3)	PW(Alt 1)
\$ 1,061,536.65	-\$ 495,853.69	\$ 93,556.21
\$ 1,661,536.65	-\$ 440,123.34	\$ 93,556.21
\$ 2,261,536.65	-\$ 384,392.99	\$ 93,556.21
\$ 2,861,536.65	-\$ 328,662.63	\$ 93,556.21
\$ 3,461,536.65	-\$ 272,932.28	\$ 93,556.21
\$ 4,061,536.65	-\$ 217,201.93	\$ 93,556.21
\$ 4,661,536.65	-\$ 161,471.58	\$ 93,556.21
\$ 5,261,536.65	-\$ 105,741.22	\$ 93,556.21
\$ 5,861,536.65	-\$ 50,010.87	\$ 93,556.21
\$ 6,461,536.65	\$ 5,719.48	\$ 93,556.21
\$ 7,061,536.65	\$ 61,449.84	\$ 93,556.21
\$ 7,661,536.65	\$ 117,180.19	\$ 93,556.21
\$ 8,261,536.65	\$ 172,910.54	\$ 93,556.21
\$ 8,861,536.65	\$ 228,640.90	\$ 93,556.21

Figure 8. Rainbow Diagram of PW vs. Annual Sales of Alternative 3



Let X = Annual Sales of Alternative 3

Annual Costs of Alternative 3:

$$\begin{aligned}
 &= \text{Operation \& Maintenance Cost} + \text{Manpower Cost (Part-time Associate } \times 1) + \text{Shrinkage Cost} \\
 &= -250694.02 - 13140 - 5307.68 \\
 &= -\$269,141.70
 \end{aligned}$$

PW of Alternative 3:

$$\begin{aligned}
 &= -112500 + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^1} + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^2} \\
 &\quad + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^3} + \frac{75000-0.17(75000)}{(1+0.1)^3}
 \end{aligned}$$

PW of Alternative 1: \$93,556.21

Computing the Break-Even Point:

Solve:

$$\begin{aligned}
 &-112500 + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^1} + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^2} \\
 &\quad + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^3} + \frac{75000-0.17(75000)}{(1+0.1)^3} = \$93,556.21
 \end{aligned}$$

$$X = \$7,225,800$$

Decision Rules based on Annual Sales of Alternative 3:

Annual Sales of Alternative 1:	Optimal Alternative:
$X \leq \$7225800$	Alternative 1
$\$7225800 \leq X$	Alternative 3

Decision Reversal based on Annual Sales of Alternative 3:

- At base value $X = \$1,061,536.65$, Alternative 1 is preferred to Alternative 3
- Change in value required for a decision reversal

$$\begin{aligned}
 &= 7225800 - 1061536.65 \\
 &= \$6,164,263.35
 \end{aligned}$$

- Hence, %-Change required for a decision reversal

$$= \frac{6164263.35}{1061536.65}$$

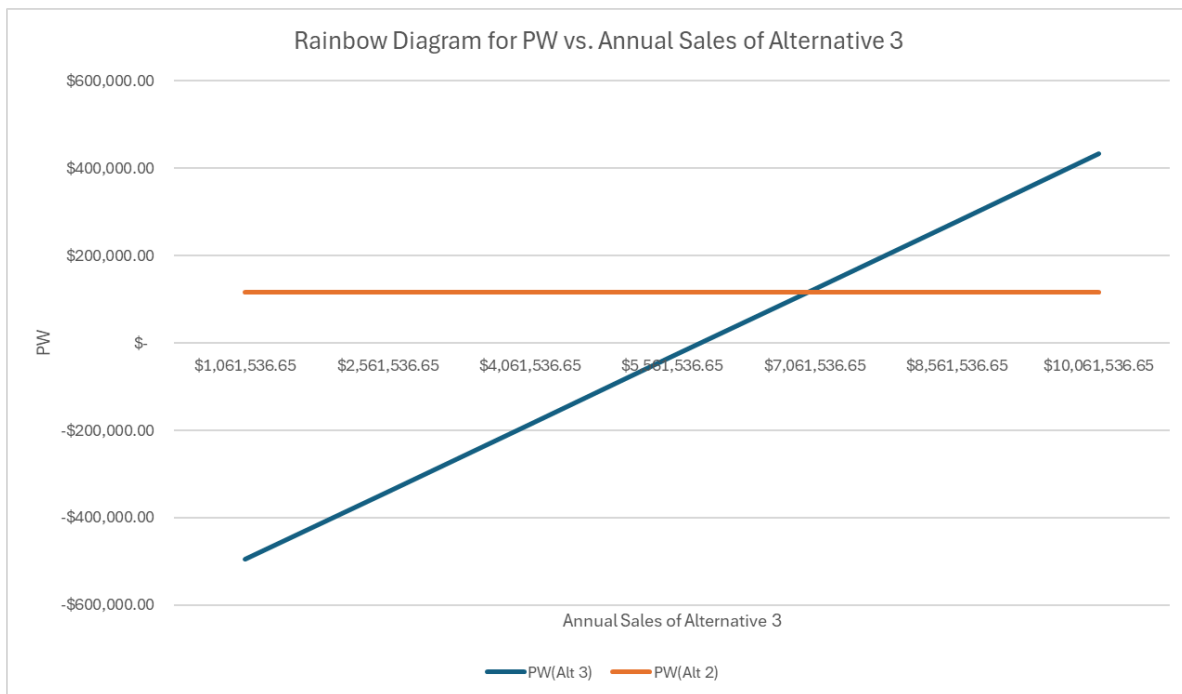
$$= + 580.69\%$$

Sensitivity Analysis on Annual Sales of Alternative 3 (Comparing Alternatives 3 and 2):

Table 6. Table of PW(Alt 3) against Annual Sales of Alt 3

Annual Sales of Alt 3	PW(Alt 3)	PW(Alt 2)
\$ 1,061,536.65	-\$ 495,853.69	\$ 117,009.83
\$ 2,561,536.65	-\$ 341,047.15	\$ 117,009.83
\$ 4,061,536.65	-\$ 186,240.62	\$ 117,009.83
\$ 5,561,536.65	-\$ 31,434.23	\$ 117,009.83
\$ 7,061,536.65	\$ 123,372.46	\$ 117,009.83
\$ 8,561,536.65	\$ 278,178.99	\$ 117,009.83
\$ 10,061,536.65	\$ 432,985.53	\$ 117,009.83

Figure 9. Rainbow Diagram of PW vs. Annual Sales of Alternative 3



Let **X** = Annual Sales of Alternative 3

Annual Costs of Alternative 3:

= Operation & Maintenance Cost + Manpower Cost (Part-time Associate x1) + Shrinkage Cost

= - 250694.02 - 13140 - 5307.68

= - 269141.70

PW of Alternative 3:

$$\begin{aligned} &= -112500 + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^1} + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^2} \\ &\quad + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^3} + \frac{75000-0.17(75000)}{(1+0.1)^3} \end{aligned}$$

PW of Alternative 2: \$117,009.83

Computing the Break-Even Point:

Solve:

$$\begin{aligned} &-112500 + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^1} + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^2} \\ &\quad + \frac{(1-0.17)(0.05X-269141.7)+0.17 \times 37500}{(1+0.1)^3} + \frac{75000-0.17(75000)}{(1+0.1)^3} = \$117,009.83 \end{aligned}$$

X = \$6,999,890.00

Decision Rules based on Annual Sales of Alternative 3:

Annual Sales of Alternative 1:	Optimal Alternative:
$X \leq \$6999890$	Alternative 2
$\$6999890 \leq X$	Alternative 3

Decision Reversal based on Annual Sales of Alternative 3:

- At base value $X = \$1061536.65$, Alternative 2 is preferred to Alternative 3
- Change in value required for a decision reversal
 $= 6999890 - 1061536.65$
 $= \$5,938,353.35$
- Hence, %-Change required for a decision reversal
 $= \frac{5938353.35}{1061536.65}$
 $= +559.41\%$

6. FURTHER ANALYSIS: CONSIDERATION OF QUALITATIVE FACTORS

Financial factors aside, retailers are also concerned with how non-financial factors of the in-store checkout processes may affect customers' shopping experience. To investigate the key qualitative factors that contribute to customers' satisfaction with in-store checkout processes, we conducted the Analytical Hierarchy Process with a limited sample size of 5, aged 21 to 40.

The scores that each respondent assigned to a factor is according to the following scale (Goepel, 2013):

1	Equal Importance: Two elements contribute equally to the objective
3	Moderate Importance: Experience and judgement slightly favour one element over another
5	Strong Importance: Experience and judgement strongly favour one element over another

7	Very Strong Importance: One element is favoured very strongly over another, its dominance is demonstrated in practice
9	Extreme Importance: The evidence favouring one element over another is of the highest possible order of affirmation

6.1. Wait Time

According to 92% of retailers, the cost of a shopper waiting in line at a physical store negatively affects company revenues (Zippin, 2023). As customers' demand for efficiency grows, businesses have reported that out of the customers they have lost, 75% of them were lost due to long wait times (Tsernov, 2017). This makes wait time a factor retailers should prioritise when selecting an optimal checkout system.

6.2. Smoothness

96% of shoppers indicate that ease of checkout are important factors when choosing where to shop (CStore Decisions, 2019). While self-checkout is favoured by some customers for its speed, 67% of customers were reported to have experienced a failure at the self-checkout lane (Meyersohn, 2022). Therefore, we would like to investigate how the smoothness and ease of each checkout compares against other alternatives.

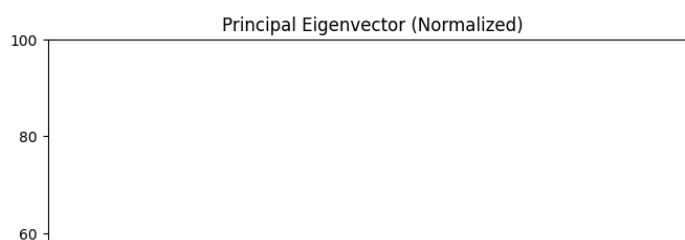
6.3. Customer Service

A 2023 survey found that average customer service scores fell by an average 1.2 points, from 2022 to 2023, to a new record low. This is reportedly attributed to staff feeling overwhelmed by the number of self-checkout machines they had to manage (Nott, 2023). Yet, customer service is significantly valued by customers - the top reason customers switch brands is because they feel unappreciated.

The sample of shoppers ranked the 3 key criteria of in-store checkout by comparing:

- Wait Time vs. Smoothness,
- Wait Time vs. Customer Service,
- Smoothness vs. Customer Service

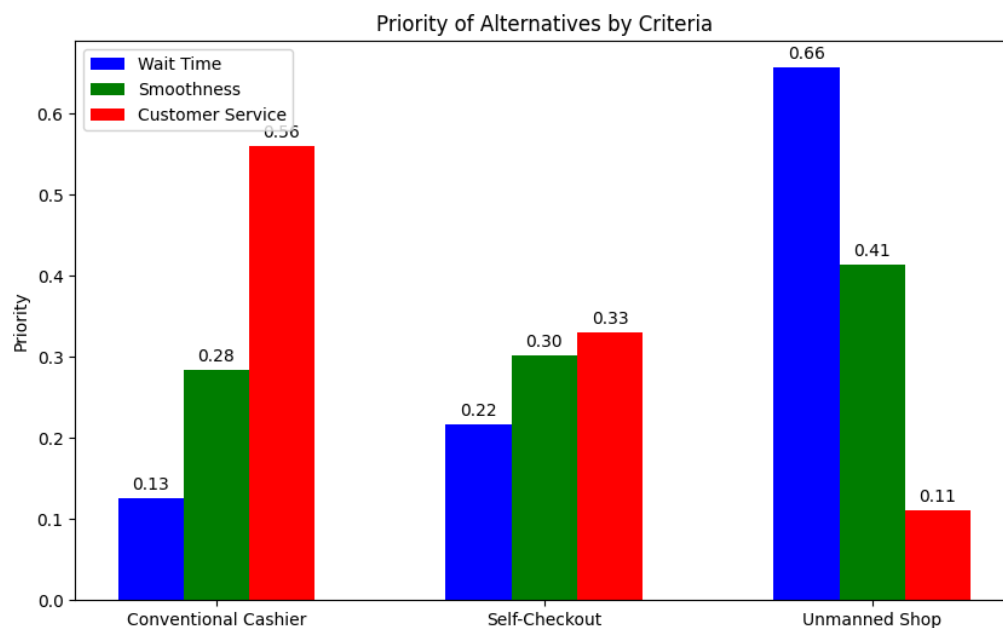
From the responses, the normalised eigenvector matrix (Appendix D) and bar chart were derived.



By analysing the normalised principal eigenvector, it can be observed that respondents value wait time the most (weight of 0.4931), followed by smoothness of checkout (weight of 0.3121) and lastly customer service (weight of 0.1949).

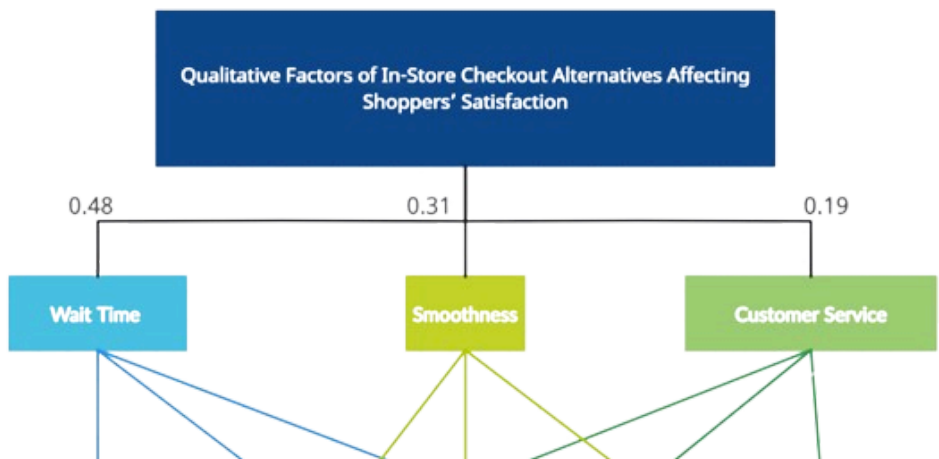
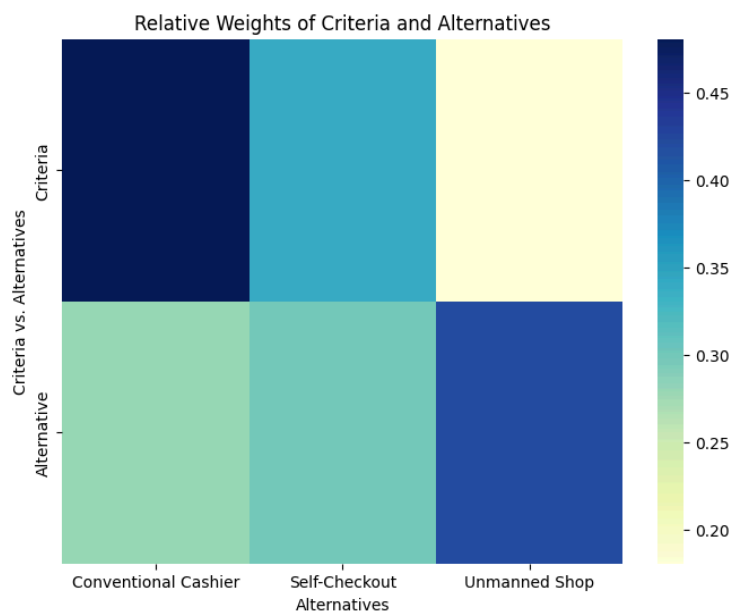
6.4. Prioritisation of Alternatives

Next, for each criterion, respondents ranked how each alternative measured against each other by scoring conventional cashier, self-checkout and unmanned shops against each other for each criteria. The eigenvector matrices (Appendix B) were derived and the following bar chart to compare how respondents perceived the qualities of each alternative.



Evident from the graph, unmanned shops are prioritised by wait time and smoothness; conventional cashiers are prioritised by customer service, but fare the poorest in wait time and smoothness. Self-checkouts are ranked moderately across all 3 criteria by respondents.

By calculating the normalised total relative weights of each alternative, the following heat map was also derived, displaying the relative weights of the criteria and the alternatives, as well as the AHP chart.



In consensus with the findings above, conventional cashier (Alternative 1) is awarded the lowest weight as it was scored the worst for the 2 criteria that customers prioritise the most: wait time and smoothness. Since self-checkout (Alternative 2) is ranked moderately across all 3 criteria, it is obvious that it will also derive the medium weight among the alternatives.

Lastly, as the unmanned shop (Alternative 3) was scored the highest for the 2 most valued criteria, it is given the greatest weight.

Notably, the difference in weights between the conventional cashier and the self-checkout is not as significant as that between the self-checkout and unmanned shop options.

Against the backdrop of the Singaporean retail landscape, the findings from the investigation of the non-quantitative factors aligns with the fact that convenience and efficiency is often prioritised by shoppers for in-store experiences.

7. RECOMMENDATIONS

Considering the results of the financial and non-financial factors, convenience shop owners are recommended to **prioritise Alternative 2 over Alternative 1**, based on the assumptions that they have sufficient budget for the initial investment required for Alternative 2, and they are not risk-averse in making this investment. Based on the results from the base case analysis and simulations, it is more probable that Alternative 2 has a higher PW over the same study period than Alternative 1. However, it carries a greater variance than Alternative 1. Furthermore, its upside potentials could range from higher to lower than that of Alternative 1, based on our simulations. Moreover, Alternative 2 performed slightly better in terms of providing customers with a pleasant in-store checkout experience, on the grounds of non-financial factors.

However, should convenience shop owners find that they are not willing or able to invest the initial costs into setting up Alternative 2, or are more risk-averse investors, Alternative 1 is still a largely desirable option as it offers a positive PW and generally lower variance and hence, risk. Although it performed the poorest by the non-financial factors, the differences in preference between Alternatives 1 and 2 are not drastically significant.

Unless shop owners are confident that the sales of their shop will increase due to unaccounted reasons (e.g. higher footfall due to newly-built facilities in the area), they are not recommended to consider Alternative 3 as a viable option.

8. CONCLUSION

While the analysis of financial and non-financial factors in this report provided contradicting results, this highlights that convenience shop owners have to be prudent in balancing customers' ideal checkout preference with the financial feasibility of operating more technologically innovative options such as a checkout-free or unmanned system (*Alternative 3*).

Also, it is important to note that the limited sample size involved in the AHP may cause results to be skewed, especially due to the age range of shoppers. Older shoppers may value customer service more and give the smoothness of self-checkout kiosks and unmanned stores lower scores. Therefore, shop owners should also consider the location of the store and the demographic it appeals to, before considering how the non-financial factors may be perceived differently by its shoppers.

As there are multiple methods and complexity of implementing the novel technology required for a checkout-free system and the benefits are still largely uninvestigated by research, the risk of failure in implementation is relatively higher. This corroborates with real-world observations where Amazon removed its Just Walk Out technology from its Amazon Fresh stores (AP, 2024) as the actual non-financial benefits perceived by customers were not as significant as predicted.

Ultimately, given the current information and assumptions, a conventional checkout system (*Alternative 1*) or a self-checkout system (*Alternative 2*) are more financially feasible options, as the costs for these technologies have been lowered over time and the certainties of implementation are guaranteed.

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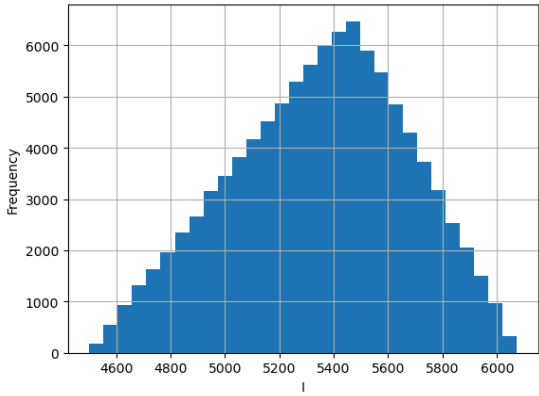
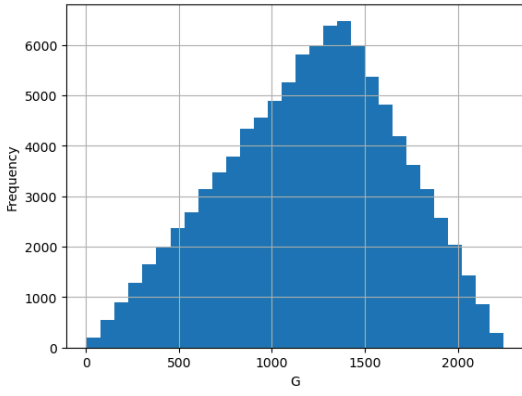
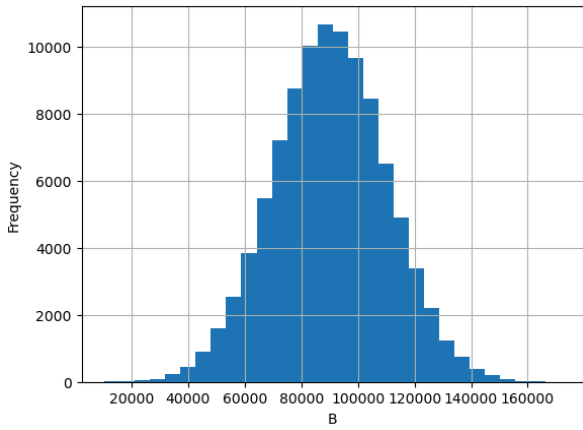
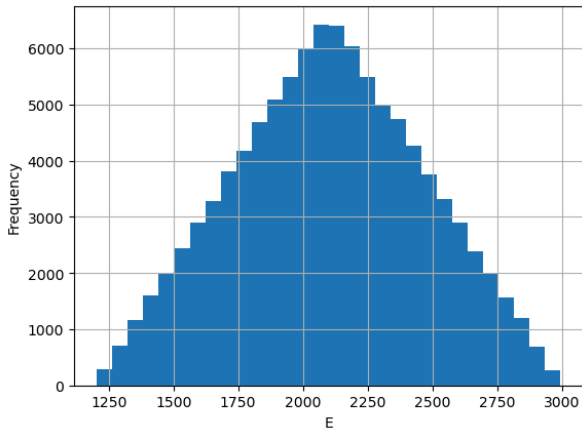
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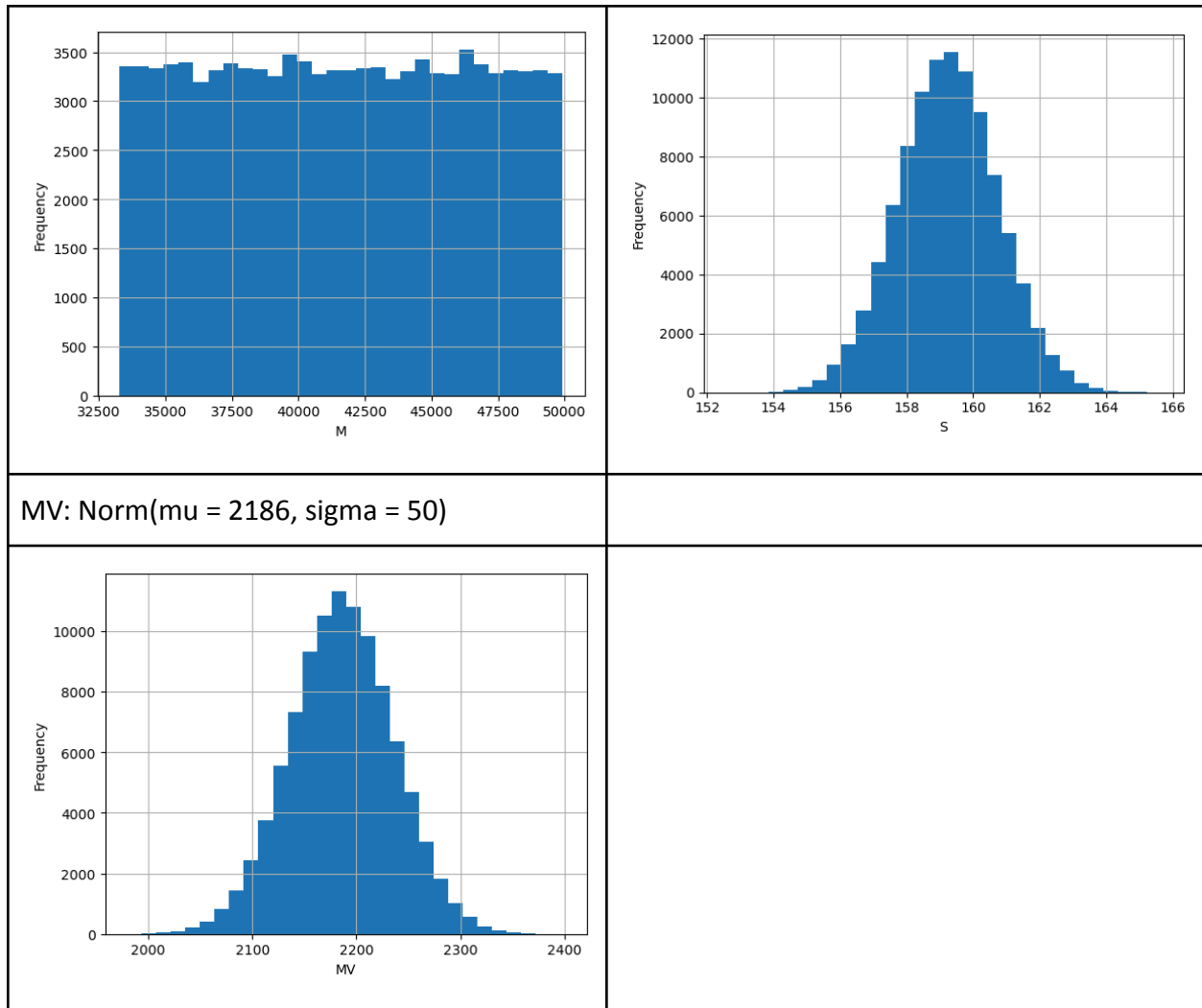
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APPENDIX

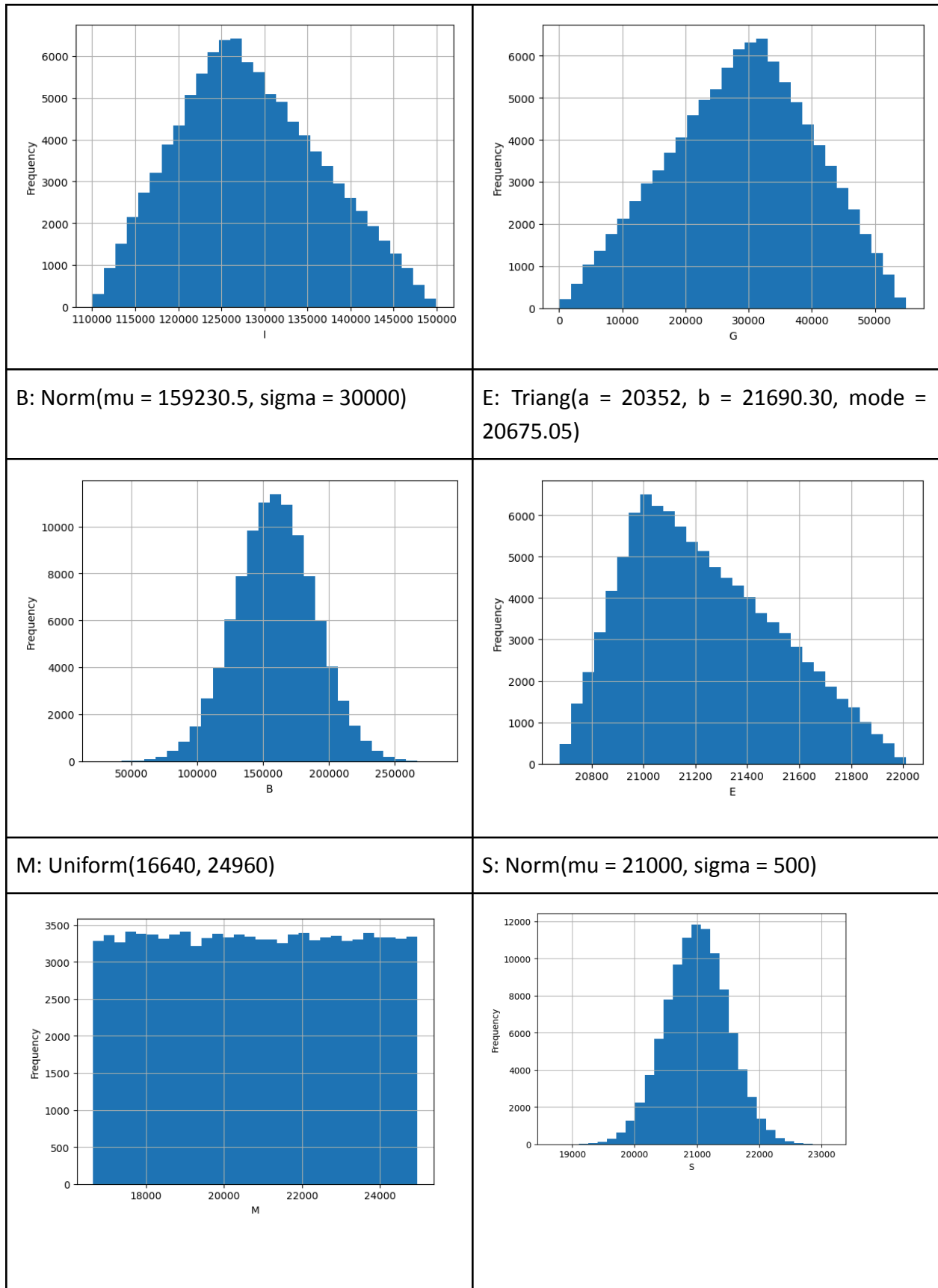
Appendix A: Distribution of PW Components for Alternative 1

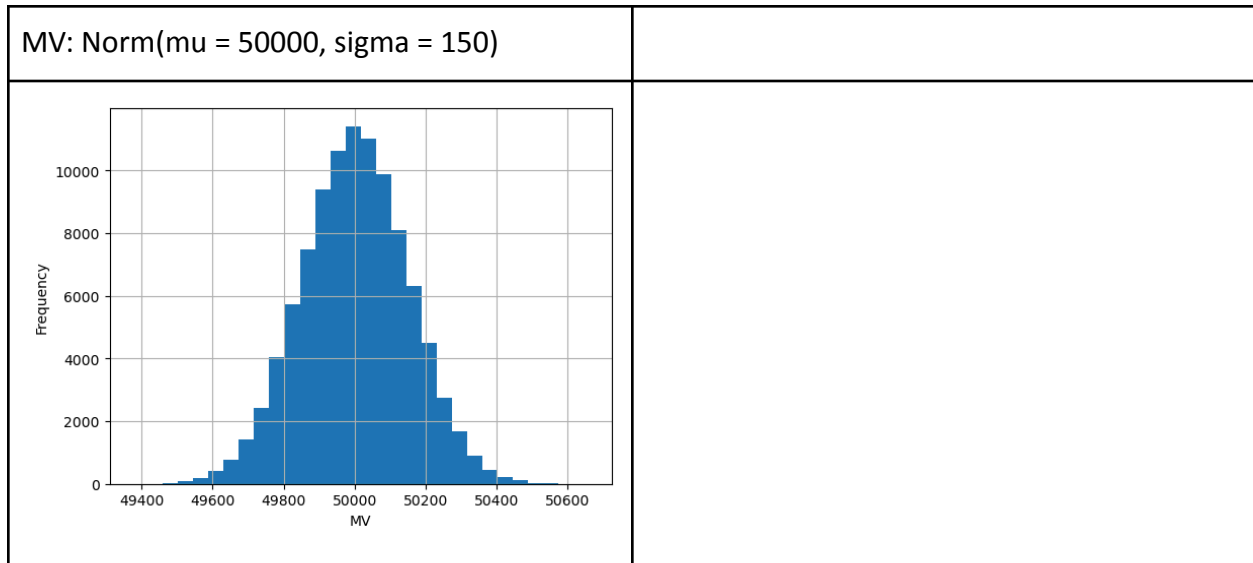
I: Triang(a = 4500, b = 6076, mode = 5466)	G: Triang(a = 0, b = 2250, mode = 1366.6)
 A histogram representing the distribution of variable I. The x-axis is labeled 'I' and ranges from 4600 to 6000 with major ticks every 200 units. The y-axis is labeled 'Frequency' and ranges from 0 to 6000 with major ticks every 1000 units. The distribution is triangular, starting near zero at 4600, peaking at a frequency of approximately 6000 around the value 5466, and tapering off to zero at 6000.	 A histogram representing the distribution of variable G. The x-axis is labeled 'G' and ranges from 0 to 2000 with major ticks every 500 units. The y-axis is labeled 'Frequency' and ranges from 0 to 6000 with major ticks every 1000 units. The distribution is triangular, starting at zero at 0, peaking at a frequency of approximately 6000 around the value 1366.6, and tapering off to zero at 2250.
B: Norm (mu = 90230.61525, sigma = 20000)	E: Triang (a = 1200, b = 3000, mode = 2100)
 A histogram representing the distribution of variable B. The x-axis is labeled 'B' and ranges from 20000 to 160000 with major ticks every 20000 units. The y-axis is labeled 'Frequency' and ranges from 0 to 10000 with major ticks every 2000 units. The distribution is bell-shaped (normal), centered around 90230.61525, with a peak frequency of approximately 10000.	 A histogram representing the distribution of variable E. The x-axis is labeled 'E' and ranges from 1250 to 3000 with major ticks every 250 units. The y-axis is labeled 'Frequency' and ranges from 0 to 6000 with major ticks every 1000 units. The distribution is triangular, starting at zero at 1200, peaking at a frequency of approximately 6000 around the value 2100, and tapering off to zero at 3000.
M: Uniform (33280, 49920)	S: Norm (mu = 159.2304975, sigma = 1.5)



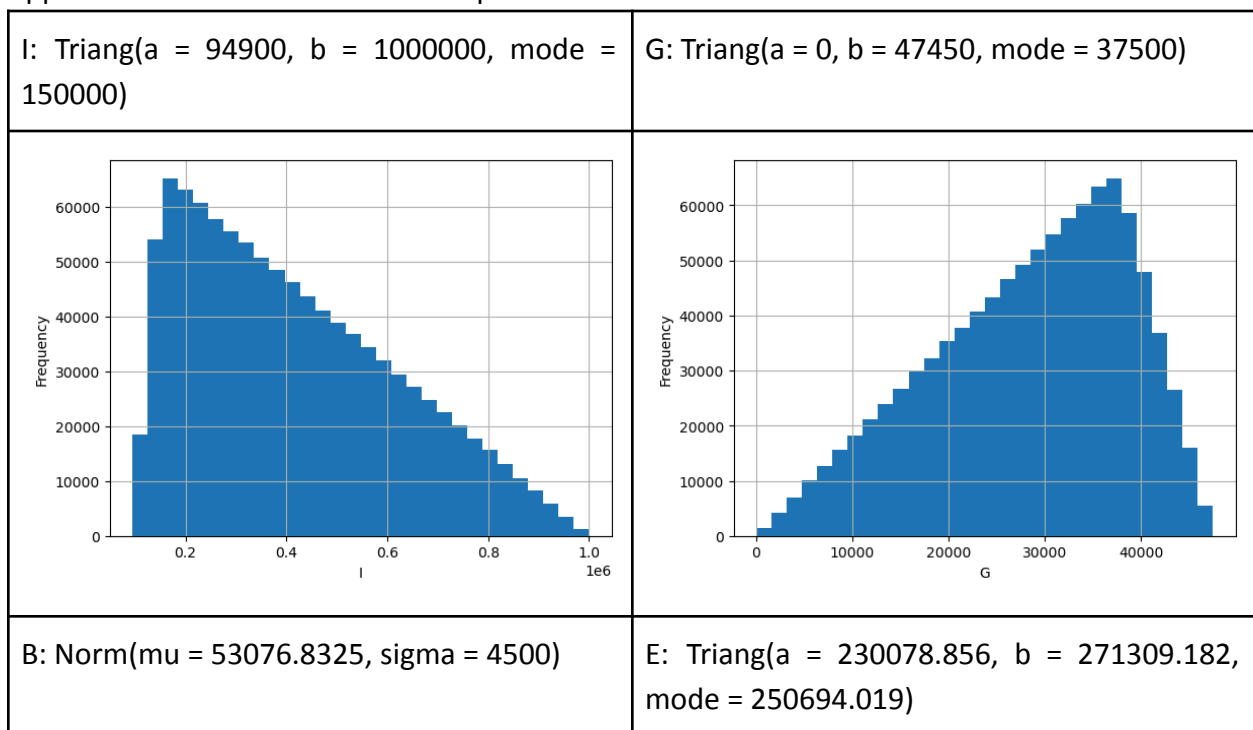
Appendix B: Distribution of PW Components for Alternative 2

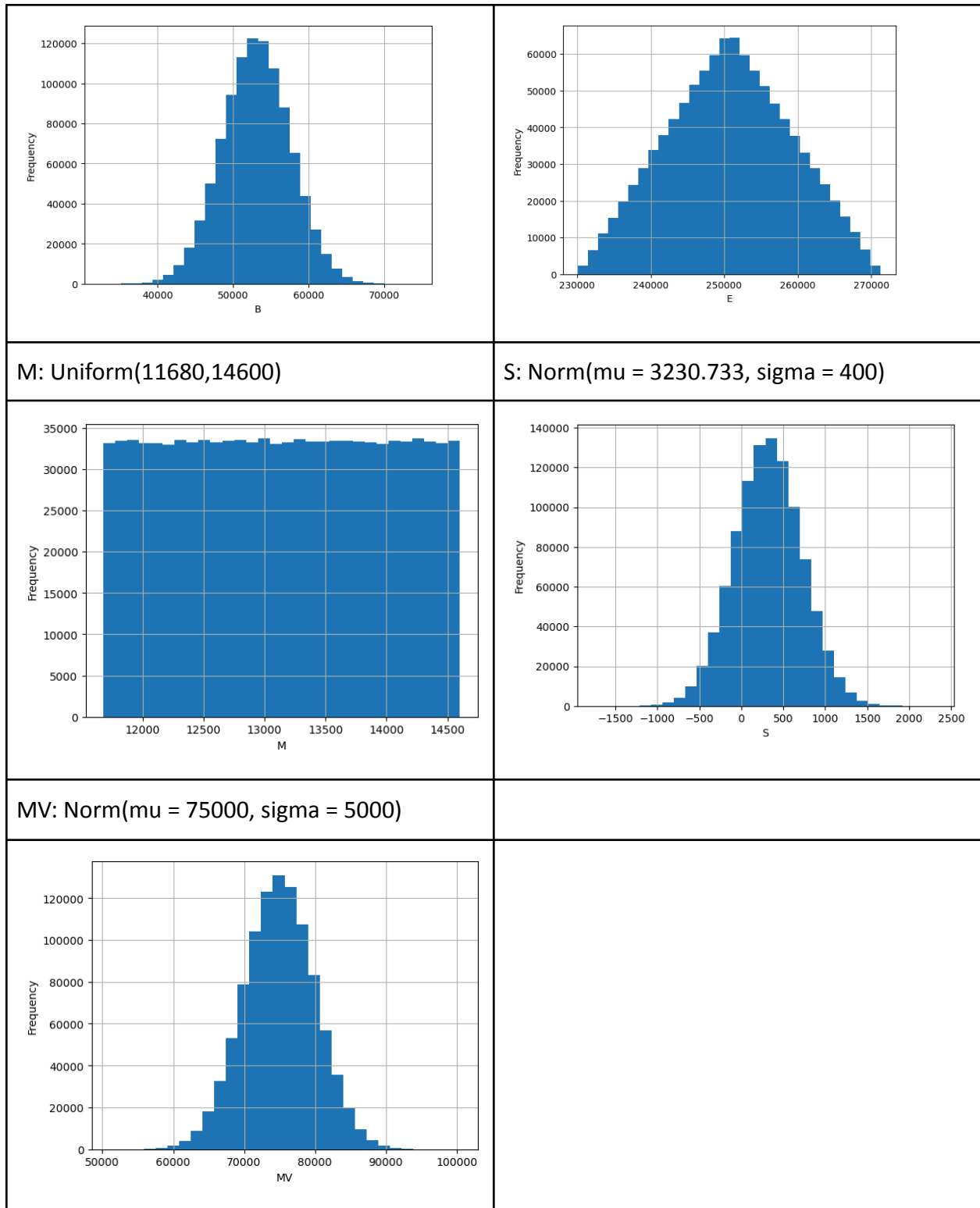
I: Triang($a = 110000$, $b = 150000$, mode = 125000)	G: Triang($a = 0$, $b = 55000$, mode = 31250)
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Appendix C: Distribution of PW Components for Alternative 3





Appendix D: Eigenvector of criteria comparison (AHP)

	Wait Time	Smoothness	Customer Service
Wait Time	1	$\frac{17}{9}$	2
Smoothness	$\frac{9}{17}$	1	$\frac{5}{2}$
Customer Service	$\frac{1}{2}$	$\frac{2}{5}$	1

Appendix B: Eigenvector of comparison of Wait Time factor among all alternatives (AHP)

	Alternative 1	Alternative 2	Alternative 3
Alternative 1	1	$\frac{1}{2}$	$\frac{2}{9}$
Alternative 2	2	1	$\frac{2}{7}$
Alternative 3	$\frac{9}{2}$	$\frac{7}{2}$	1

Eigenvector of comparison of Smoothness factor among all alternatives (AHP)

	Alternative 1	Alternative 2	Alternative 3
Alternative 1	1	$\frac{7}{9}$	$\frac{5}{6}$
Alternative 2	$\frac{9}{7}$	1	$\frac{3}{5}$
Alternative 3	$\frac{6}{5}$	$\frac{5}{3}$	1

Eigenvector of comparison of Customer Service factor among all alternatives (AHP)

	Alternative 1	Alternative 2	Alternative 3
Alternative 1	1	$\frac{12}{7}$	$\frac{5}{6}$
Alternative 2	$\frac{7}{12}$	1	3

Alternative 3	$\frac{6}{5}$	$\frac{1}{3}$	1
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